8 January 1973
Superseding
MIL-N-81207(AS)
1 November 1968

MILITARY SPECIFICATION

NAVIGATIONAL SET, TACAN AN/ARN-84(V)

This specification has been approved by the
Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide, when installed in aircraft, bearing, distance and tone identity information to indicate the location of a complementary surface station with respect to the aircraft; the equipment shall indicate the distance and bearing to a properly equipped aircraft; and the equipment shall indicate range and inverse bearing to appropriate beacons. This equipment is one airborne portion of the short range air navigation system known as TACAN (Tactical Air Navigation).

1.2 Classification - The equipment covered by this specification shall consist of the following items:

Item	Type Designation	Appl. Para.
Receiver-Transmitter, TACAN	RT-1022/ARN-84(V)	3.5.1
or		
Receiver-Transmitter, TACAN	RT-1127/ARN-84(V)	3.5.2
Control, Navigation Set	C-8734/ARN-84(V)	3.5.3
or		
Control, Navigation Set	C-9054/ARN-84(V)	3.5.4
or		
Control, Navigation Set	C-9475/ARN-84(V)	3.5.5
or		
Control, Navigation Set	C-9483/ARN-84(V)	3.5.6

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Converter, Signal Data	CV-2837/ARN-84(V)	3*5.7
or		
Converter, Signal Data	CV-3135/ARN-84(V)	3.5.8
or		
Converter, Signal Data	CV-3134/ARN-84(V)	3*5.9
or		
Converter, Signal Data	CV-3136/ARN-84(V)	305.10
Base, Shockmount, Electrical Equipment	MT-4354/ARN-84(V)	3.5.11
or		
Base, Hardmount, Electrical Equipment	MT-4443/ARN-84(V)	3.5.12
or		
Base, Shockmount, Electrical Equipment	MT-4616/ARN-84(V)	3.5.13
or		
Base, Shockmount, Electrical Equipment	MT-4615/ARN-84(V)	3.5.14
or		
Base, Shockmount, Electrical Equipment	MT-4617/ARN-84(V)	3.5.15

103 Associated Equipment - This equipment shall operate with the associated equipment listed in 6.7 to the extent specified herein.

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2. APPLICABLE DOCUMENTS

2.1 General - The following documents form a part of this specification to the extent specified herein. Listed are the issues of the documents in effect for the previous procurement. However, in lieu of the issue listed, the contractor shall use the latest issue in effect of these documents where feasible. If the use of the latest issue will affect design performance, or interchangeability of any replaceable part, then the issue of the document listed below shall be used.

SPECIFICATIONS

Military

MIL-C-172C Supplement-1D 8 Dec. 1958	Cases, Bases, Mounting, and Mounts. Vibration
MIL-W-5088C 26 May 1965	Wiring; Aircraft, Installation of
MIL-E-5400L 1 August 1969	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422E 13 April 1959 And Amendment-2 15 Nov. 1961	Testing, Environmental, Aircraft Electronic Equipment
MIL-C-5824A 11 March 1952	Indicator, Course ID-250A/ARN
MIL-C-6781B 13 Sept. 1960	Control Panel: Aircraft Equipment, Rack or Console Mounted
MIL-P-7788D 14 April 1967	Plate, Plastic, Lighting
MIL-M-7793C 14 August 1961 Amendment-2 30 June 1967	Meter, Time Totalizing, General Specification for
MIL-I-9229 4 Sept. 1953 Amendment-2 9 Sept. 1965	Indicator, Course ID-387/ARN

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MIL-E-17555G 3 July 1968 Amendment-1 10 Jan. 1969	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303B 1 Sept. 1966	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307 15 Sept. 1969	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment Including Ground Support Equipment
MIL-I-18955 1 Oct. 1955	Indicator, Range ID-388/APN
MIL-S-20708C 1 April 1965 Amendment-2 27 August 1969	Synchros, General Specification for
MIL-I-22075A 1 Oct. 1965	Indicator, Bearing-Distance-Heading ID-663/U
MIL-I-23366A 28 Oct. 1966 Amendment-1 15 Jan. 1968	Indicator, Horizontal Situation ID-1013/A
MIL-A-25708A 13 March 1964	Antenna AT-741/A
MIL-S-25879B 12 Feb. 1968	Switch, Radio Frequency Transmission Line SA-521/A
MIL-I-27848A 25 March 1966	Indicator, Horizontal Situation AQU-4/A
MIL-D-81170	Data Link Set AN/ASW-27
MIL-M-81288 1 Jan. 1966 And Amendment-1 15 July 1968	Mounting Bases, Flexible Plastic Foam

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Naval Air Systems Command

AR-5 1 Oct. 1966	Microelectronic Devices Used in Avionics Equipment, Procedures for Selection and Approval of
AR-8A 11 March 1969	Versatile Avionic Shop Test System, Avionic System Compatibility General Requirement for
AR-9 1 May 1967 Amendment-1 5 April 1968	Versatile Avionic Shop Test System Test Program, General Requirements for
AR-10A 1 Jan. 1969	Maintainability of Avionic Equipment and Systems, General Requirements for
AR- 34 26 March 1969	Failure Classification for Reliability Testing, General Requirements for

STANDARDS

Military

MIL-STD-291B 13 Dec. 1967	Standard Tactical Navigation (TACAN) Signal
MIL-STD-461A 1 August 1968 Notice-2 20 March 1969 Notice-3 1 May 1970	Electromagnetic Interference Characteristics, Requirements of Equipment
MIL-STD-462 31 July 1967 Notice-1 1 August 1968 Notice-2 1 May 1970	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704A 9 August 1966 And Notice-1 7 Feb. 1968	Electric Power, Aircraft, Characteristics and Utilization

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MIL-STD-781B 15 NOV. 1967 Notice 1 28 July 1969	Reliability Tests, Exponential Distribution
MIL-STD-785 28 March 1969	Reliability Program for Systems and Equipment Development and Production
MIL-ST-794B 11 March 1969	Parts and Equipment, Procedures for Packaging and Packing
MS 17322D 30 June 1967	Meter, Time Totalizing Miniature, Digital 115 Volts 400 Cycle

2.1.1 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Comanding officer, Publications and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

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3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Microelectronic technology shall be considered and microelectronic items shall conform to requirements specified herein.

(2) other parts and materials requirements shall conform to Specification MIL-E-5400.

(3) Nonrepairable subassemblies shall be used in accordance with Specification AR-10 and as outlined in Specification MIL-E-5400.

(4) When previously produced models of this equipment did not use nonrepairalble subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

(5) Repairable subassemblies shall not use greases to seal them or to reduce internal arcing. Such sealing shall be by O-rings or other devices not subject to contamination by dust and grit.

(6) Plug-in subassemblies shall be so designed as to preclude improper interchange or insertion.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials (including electron tubes, transistors and diodes) other than microelectronic devices shall be obtained as outlined in Specification MIL-E-5400. Microelectronic devices shall be approved as outlined in AR-5.

3.2.2 Microelectronic Modular Assemblies - When used, Microelectronic Modular Assemblies shall meet the requirements of Specification AR-5. Conformal coatings, encapsulant, embedments

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or potting materials used with modular assemblies containing integrated circuits and discrete parts shall be easily removable without damage to the assembly.

3.2.3 Modules - The electronic portions of the equipment shall be functionally modularized in accordance with Specification AR-10.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and Workmanship, except as otherwise specified herein.

3.3.1 Total weight - The total weight of the equipment, excluding cables and blower, shall be a minimum consistent with good design and shall not exceed 54 pounds.

3.3.2 Reliability - The contractor shall conduct a reliability program using MIL-STD-785 as a guide. On a reorder from a supplier who has previously produced the equipment, the program previously used may be continued unless otherwise indicated in the contract or order.

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 1,000 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a total operating life of 10,000 hours with reasonable servicing the replacement of parts. Parts requiring scheduled replacement shall be-specified by the contractor.

3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The equipment, including any Built-In-Test provisions, shall have at least 500 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.3.2.4 Time Totalizing Meter - The following units shall contain a time totalizing meter in accordance with Specification. MIL-M-7793.

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Unit	Type of Meter
RT-1022/ARN-84(V)	MS 17322
CV-2837/ARN-84(V)	MS 17322
RT-1127/ARN-84(V)	MS 17322
CV-3134/ARN-84(V)	MS 17322
CV-3135/ARN-84(V)	MS 17322
CV-3136/ARN-84(V)	MS 17322

3.3.3 Cabling and Connections

3.303.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.3.3.2 Interconnections Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to need interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.4 Control Panels - All rack ox console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all panels must be approved by the procuring activity prior to preproduction testing.

3.3.5 Interchangeability

3.3.5.1 Electrical and Mechanical - The equipment shall meet the interchangeability requirements of Specification MIL-E-54000

3.3.5.2 Adjustments - There shall be no adjustments to permit interchange of modules.

3.3.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall meet the requirements of MIL-STD-461, except as modified herein.

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3.3.6.1 Radiated and Conducted Interference - The antenna conducted outputs in T/R and A/A modes shall meet the requirements for spurious and harmonic emanations as required by the CE06 test limits, except that the antenna conducted spurious CW power shall not exceed -47 dBm at frequencies between 950 MHz and 1250 MHz in the REC mode or between pulse pairs in the T/R and A/A modes.

3.3.6.2 Low Frequency Conducted Interference - The conducted interference requirements of MIL-STD-461 shall be modified as follows:

CE01 - increase allowable interference by 30 db for the first three odd harmonics of 400 Hz and by 20 db for higher harmonics

CE01 - increase allowable interference by 10 db from 14 KHz to 20 KHz

3.3.7 Provisions for Maintainability - Built-in test equipment, construction and packaging, provisions for test points, and other maintainability parameters shall be as specified in Specification AR-10.

3.3.7.l Fault Location Indicators - The fault location indicators required by AR-10 shall indicate a malfunction to the faulty WRA.

3.3.7.2 Compatibility with VAST - The equipment shall be compatible with the Versatile Avionic Shop Test System (VAST) and shall meet the requirements of Specification AR-8. When required by contract, VAST Operational Test Program Sets shall be furnished *in* accordance with Specification AR-9. If Operational VAST Test Program Sets exist for the equipment, and changes to the equipment are made which will affect the fault diagnosis procedure, changes to the existing Test Program Sets shall be prepared as part of the equipment changes in accordance with Specification AR-9.

3.3.8 Nomenclature, Nameplates and Identification Marking -Nomenclature and serial number assignment, nameplate approval and identification marking shall be in accordance with Specification MIL-N-18307.

3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

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Temperature	Room ambient (25°C±5°C)
Altitude	Normal Ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115±1.0 VAC, 27.5±0.5 VDC, and 26.0±0.5 VAC

3.3.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 2X equipment, except as modified herein.

3.3.10.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves IV and II of Specification MIL-E-5400. The Navigation Set Controls shall conform to Curve I except the amplitude shall not exceed 5g.

3.3.11 Warm-Up Time - The time required for the equipment warm-up prior to operation shall be kept to a minimum and shall not exceed 2 minutes under standard conditions and 6 minutes at extreme service conditions.

3.3.12 Input Electrical Power

3.3.12.1 Operating Power - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources having characteristics and limits as defined in MIL-STD-704. The power required shall not exceed the specified amounts.

(1) AC Power (Single Phase), Category B

115 volts,	REC Mode	300 watts
	T/R Mode	300 watts
	A/A Mode	335 watts
	BCN Mode	350 watts

(2) DC Power, 28V, Category B, 3.2A

3.3.12.2 Lighting Power - Input power for lighting shall require not more than 8 watts at 28 volts DC for the C-8734/ARN-84(V), not more than 4 watts at 5 volts DC for the C-9054/ARN-84(V) or for the C-9475/ARN-84(V) or for the C-9483/ARN-84(V).

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3.3.12.3 Blower Power - Provisions shall be made to supply up to 1.5 amperes at 115 VAC, 400 Hz, to power an external cooling air source when the equipment is energized.

3.3.12.4 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment.

3.3.13 Cooling - The equipment shall operate satisfactorily when supplied with cooling air from the internal blower and aircraft supplied conditioned air as described in Figure 1.

3.3.14 Acoustic Noise - No malfunction of the equipment shall occur when subjected to a noise of 120 dB relative to 0.0002 dyne per square centimeter at frequencies between 20 and 10,000 Hz.

3.4 Performance - Unless otherwise specified, values set forth to extablish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Operation - The equipment, when properly interconnected shall transmit to and receive from an aircraft with an AN/ARN-21, AN/ARN-52(V), AN/ARN-72, AN/ARN-84(V) , AN/ARN-90, or AN/ARN-91 TACAN Set installed, or from a complementary navigation beacon, a combination of pulsed signals from which it shall derive and process the information necessary to perform the functions as described in Table I. Furthermore, the equipment, independently of the signals received from the beacon or complementary aircraft, shall provide a blanking pulse output during the time of its own transmission in the TRANSMIT-RECEIVE and AIR-TO-AIR modes which will cause blanking of other equipment operating in the same frequency band.

3.4.2 Characteristics of Received TACAN Signals - This equipment shall operate and provide the accuracy requirements specified herein when receiving pulse code and pulse amplitude type signals of a surface or air facility conforming to MIL-STD-291 and in addition the following:

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3.4.2.1 Pulse Rise and Fall Time - The time required for the leading edge of the pulse to rise from 10 to 90 percent of its maximum voltage amplitude shall be 2.5 ± 0.5 microsecond. Correspondingly the fall time shall be 2.7 ± 1.3 microseconds.

3.4.2.2 Pulse Coding (Mode X Channels) - The pulses shall be coded in pairs with a spacing which shall be 12 ± 0.5 microsecond as measured at the 50 percent maximum voltage amplitude points from the leading edge of the first pulse of the leading edge of the second pulse. The X mode A/A interrogating pulse spacing shall be 12.0 ± 0.5 microseconds.

3.4.2.3 Pulse Coding (Mode Y Channels) - The equipment shall be capable of using ground reply pulses, coded in pairs with a spacing of 30 ± 0.5 microsecond as measured at the 50 percent maximum voltage amplitude points from the leading edge of the first pulse of the leading edge of the second pulse. The Y mode AIR-TO-AIR interrogating pulse spacing shall be 24 ± 0.5 microseconds.

3.4.2.4 Squitter Distribution - The received squitter may be reasonably even, not necessarily random, but asynchronous with modulation frequency and with DME interrogations.

3.4.2.5 Frequency Variations - The received signal frequency may vary from the nominal RF channel. frequency by ±.0087 percent.

3.4.2.6 Antenna Rotation Frequency - Performance shall be maintained for rotation frequencies of 15 Hz ± 0.8%. This tolerance shall also apply for the ninth harmonic thereof.

3.4.3 Frequency Characteristics - The equipment shall transmit and receive radio frequency signals on frequencies as specified in MIL-STD-291.

3.4.4 Pulse Repetition Rate - When the equipment is interrogating a beacon or ano ther aircraft, the maximum pulse repetition rate shall be 150 pulse pairs per second (pp/s) in the search mode and 30 pp/s in the track mode. In the AIR-TO-AIR mode the minimum track interrogation rate shall he 22 pp/s. The pulse repetition rate shall vary or jitter enough to preclude synchronization of transmitted pulses among any of 100 equipments on the same channel for more than 1 second in 1 hour. A search repetition rate of 135 pp/s shall be avoided to preclude synchronization with the fine bearing information.

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3.4.5 Transmitter Spectrum - In both the search and track conditions, 94% of the transmitted energy shall be contained within a 500 KHz band centered on the nominal-frequency of the assigned channel. All lobes 1 MHz or greater from the center frequency of the spectrum shall be at least 45 dB below the peak power output.

3.4.6 Effect of Transmitter Power on Other Functions Under no circumstances shall the equipment lock on to its own transmission. No degradation of any of the other functions of the equipment shall occur as a result of its own transmissions.

3.4.7 Traffic Capacity in A/A Mode - With the equipment operating in the A/A mode on a given channel, it shall be possible to transpond effectively to each of five interrogating aircraft when their equipments are set 63 channels from the given channel and when their signal strength at the antenna input connector of the equipment specified herein are within 12 to 15 dB of each other when the level of the weakest signal is any value from -90 dBm to -12 dBm. The equipment specified herein shall interrogate and lock-on to one of the five, but it shall not be required to determine the one to which it is locked-on. The equipment shall not be damaged when subjected to any number of interrogations.

3.4.8 Accuracy - All accuracies shall be realized when the equipment is properly loaded and the equipment is operated under any of the service conditions, or reasonable combination thereof, as specified in MIL-E-5400.

3.4.8.1 Distance Measurement Accuracy - The distance measurement error of the equipment shall not exceed those of Table II.

TABLE II. Distance Measurement Accuracy

SIGNAL STRENGTH	0 TO 50 NMI		> 50 NMI	
	DIGITAL	ANALOG	DIGITAL	ANALOG
-90 dBm to 0 dBm	±0.05 NMI	±0.10 NMI	±0.10 NMI	±0.15 NMI
Below -90 dBm	±0.20 NMI	±0.25 NMI	±0.20 NMI	±0.25 NMI

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3.4.8.2 Bearing Measurement Accuracy

3.4.8.2.1 Digital Bearing Accuracy - The equipment shall have bearing errors no greater than shown below, under static tracking conditions, when both 15 Hz and 135 Hz information are present.

Bearing Accuracy	Signal Strength
±0.5°	-82 dBm to 0 dBm
±1.5°	-90 dBm to -82 dBm
±2.0°	below -90 dBm

The increase in error under dynamic tracking conditions shall not exceed an additional 1° at a bearing rate of change of 5° /second. The above shall apply at all bearings and within the modulation limits of 15% to 55% with modulation indexes not more than 0.30 nor less than 0.07.

3.4.8.2.2 Analog Bearing Accuracy - The equipment shall have bearing errors no greater than shown below, under static tracking conditions, when both 15 Hz and 135 Hz information are present:

Bearing Accuracy	Signal Strength
±1.0°	-82 dBm to 0 dBm
±2.0°	-90 dBm to -82 dBm
±2.5°	below -90 dBm

The increase in error under dynamic tracking conditions shall not exceed an additional 1° at a bearing rate of change of 5° /second. The above shall apply at all bearings and within the modulation limits of 15% to 55% with modulation indexes not more than 0.30 nor less than 0.97.

3.4.8.2.3 15 Hertz Digital Bearing Accuracy - The equipment shall have bearing errors no greater than shown below, under static tracking conditions, when operating with 15 Hz information only. This requirement shall be applicable over a 7% to 40% range of modulation, with and without auxiliary reference groups.

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Bearing Accuracy	Signal Strength
±2.0°	-82 dBm to 0 dBm
±2.5°	below -82 dBm

3.4.8 .2.4 15 Hertz Analog Bearing Accuracy - The equipment shall have bearing errors no greater than shown below, under static tracking conditions, when operating with 15 Hz information only. This requirement shall be applicable over a 7% to 40% range of modulation, with and without auxiliary reference groups.

Bearing Accuracy	Signal Strength

3.4.8.3 Data Sampling Point - Distance and hearing timing information shall be derived from the leading edge of the first pulse of each received pulse pair in the REC and T/R modes, or from the leading edge of the received single pulses in the A/A mode.

3.4.9 Stability - The receiver-transmitter frequency stability shall be as follows:

3.4.9.1 Transmitter Frequency Stability - The frequency of the transmitter shall be maintained within ±0.0025% of the specified channel frequency.

3.4.9.2 Receiver Stability - The frequency of the receiver local oscillator shall be maintained within ±0.002% of its nominal value.

3.4.10 Automatic Gain Control

3.4.10.1 Automatic Gain Control in A/A Mode - In the A/A mode, the automatic gain control shall be such that the equipment shall not lock on to any reply which is more than 12 dB weaker than any other reply.

3.4.10.2 Automatic Gain Control in T/R Mode - When operating with a ground beacon generating signals in accordance with MIL-STD-291, the response of the automatic gain control to a 60 dB step function change in signal level shall be such that all

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normal TACAN responses are within specified limits within 2 seconds after the step function.

3.4.11 Blanking Pulses

3.4.11.1 Internally Generated Blanking Pulse - The receiver-transmitter shall generate a single blanking pulse, with characteristics specified below, for each interrogation pulse pair transmitted and for each single pulse reply transmitted. The blanking pulse shall be capable of suppressing other pulse equipments in the same frequency range, including another equipment identical to the one described by this specification.

(1) Positive polarity

(2) Not less than 15 nor more than 70 volts amplitude across 500 ± 50 ohms in parallel with 1,000 pf load impedance.

(3) Not less than 20 volts per microsecond rate of pulse rise (10% to 90% of pulse amplitude) .

(4) Not less than 10 volt-s per microsecond rate of pulse decay (90% to 10% of pulse amplitude) .

(5) 50 ± 3 microseconds duration at the 15 volt level.

(6) The 15 volt level on the leading edge shall occur sooner than the 10% amplitude point on the leading edge of the corresponding transmitted reply pulse by an amount not less than 0.8 nor more than 5.0 microseconds.

(7) Any spurious output at the blanking pulse terminal shall be not more than one volt in amplitude.

(8) The application of the internally generated blanking pulse to its own receiver shall be consistent with the receiver recovery times necessary for operation in both the X and Y modes.

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3.4.11.2 Blanking From External Source - The equipment shall provide for blanking its own receiver during the transmission time of other pulse equipments in the same frequency range when a blanking pulse from an external equipment is applied to the receiver-transmitter. The blanking pulse characteristics will be the same as those defined in 3.4.11.1, except that the blanking pulse duration shall be long enough to effectively bracket the transmission time.

3.4.12 Transmitter Muting - The transmitter shall be automatically muted during channeling in the T/R and A/A modes, when switching between the T/R and A/A modes, and when switching between mode X and mode Y.

3.4.13 Radio Interference

3.4.13.1 On Frequency - With the equipment tracking on a signal of -80 dBm, the specified performance shall not be affected by-the following type and level of interference.

Type	Level
6,000 randomly spaced single pulses per second each 3.5 micro-seconds wide in REC, T/R, and A/A ranging.	-20 dBm

3.4.13.2 Off Frequency Interference

(1) Performance with no desired signal present shall not be affected and the equipment shall remain in search when CW signals of the following frequencies and amplitudes are introduced at the antenna terminal:

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Frequency	Amplitude
0.1 MHz to 950 MHz	-30 dBm
Intermediate Frequencies	-10 dBm
1250 MHz to 10,000 MHz	-30 dBm
950 MHz to 1,250 MHz except within 3 MHz of the desired frequency.	-30 dBm

(2) With the equipment tracking a TACAN signal at a level of -80 dBm the specified performance shall not be degraded by the introduction of the following frequencies and amplitudes at the antenna terminals:

Frequency	Amplitude
0.1 MHz to 950 MHz	-30 dBm
Intermediate Frequencies	-10 dBm
1250 MHz to 10,000 MHz	-30 dBm
950 Mhz to 1,250 MHz except within 10 MHz of the desired frequency.	-30 dBm
Between 3 MHz and 10 MHz of the desired frequency	-55 dBm

3.4.13.3 On-Channel Tacan Signals - The application of on-channel Tacan signals to the input of the receiver-transmitter at any level up to +20 dBm shall not damage or adversely affect the operation or reliability of the equipment.

3.4.14 Selectivity

3.4.14.1 Adjacent and Other Channels - The equipment shall have greater than 55 dB rejection to any beacon signals complying to MIL-STD-291, on any channel other than the channel to which the equipment is turned. This requirement applies both in the presence or absence of a desired beacon signal on the selected frequency channel. In the presence of a desired signal, the undesired signal shall be considered rejected if all the performance requirement of this specification are met and no bearing errors result from the introduction of the interfering

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signal. In the absence of a desired signal, the undesired signal shall be considered rejected if no distance, bearing or identity tone information is obtained.

3.4.14.2 Co-Channel - The equipment shall be capable of rejecting the weaker of two beacon signals received on the same channel when the difference in the signal strength is 8 dB or more. The performance obtained from the stronger signal shall meet all requirements of this specification except that an additional one degree bearing error is permitted when only 8 dB of difference exists between the two signals. The 8 dB of difference. shall be measured using test signals, each having the same modulation indexes, but not exceeding a total of 40% modulation.

3.4.15 Operational Range - The operational range of the equipment shall be at least 300 ± 3 nautical miles. The equipment shall be required to perform all functions specified herein with the prescribed accuracies while operating in its respective modes within the operational range.

3.4.16 Pulse Code Discrimination - The sensitivity of the receiver shall not decrease more than 3 dB when receiving a pulse pair spaced ± 0.5 microsecond from nominal, and loss of receiver sensitivity shall be at least 70 dB for a pulse pair spacing of at least ± 4.0 microseconds from nominal.

3.4.17 Channel Selection Time - The time to switch from full sensitivity and power output on one channel to full sensitivity and power output on any other channel shall not exceed one second, under manual control.

3.4.18 Normal TACAN System Delay - The equipment shall effect an indication of 0 ± 0.05 nautical mile in the T/R mode when:

(a) The delay at the 50% leading edge amplitude point of the first RF interrogation pulse and the first reply pulse in a coded pair, is 50 ± 0.1 microseconds for X mode operation and 56*± 0.1 microseconds for Y mode operation. (*With provisions for 74 us.)

(b) The delay is measured at the antenna input terminal.

3.4.19 Direction of Search

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3.4.19.1 Distance Search Direction - When the equipment electronically scans the range in search of a reply signal the direction of scanning shall be outward in distance from zero miles.

3.4.19.2 Bearing Search Direction - In the absence of a usable bearing signal, the equipment shall cause the bearing indicator to search in the counterclockwise direction.

3.4.20 Speed of Response

3.4.20.1 Distance Speed of Response - It shall take no longer than 1 second to search, track and display the digital transponder distance signals when the transponder reply efficiency is 70%. Additional time for analog display shall be less than 3 seconds.

3.4.20.2 Bearing Speed of Response - It shall take no longer than 3 seconds to search, track and display the proper digital bearing signals. Additional time required for analog readout shall be less than 3 seconds. This shall apply to combined 15 and 135 Hz as well as 15 Hz only bearing.

3.4.21 Tracking Rate

3.4.21.1 Distance Tracking Rate - When operating with a transponder or beacon simulator which is only 50% efficient in replying to interrogations, the equipment shall be capable of locking-on and continuously tracking distance replies while moving at relative speeds up to 4,500 knots in either direction along the transponder radial.

3.4.21.2 Bearing Tracking Rate - The equipment shall be capable of locking on and maintaining track in bearing during a rate of change in either direction of 20 degrees per second.

3.4.21.3 External Phase Shift - The equipment shall be capable of locking-on and maintaining lock with a bearing rate of change of 5° per second when the phase difference between the 15 Hz and 135 Hz components of the modulation envelope external to the equipment is ±15° (at 15 Hz).

3.4.22 Memory

3.4.22.1 Distance Memory - When the equipment has been tracking distance replies for at least 10 seconds, distance

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display shall be sustained for a memory period of 8 to 12 seconds after signal loss. Memory shall be of the velocity type. If the tracked signal returns during the memory period, tracking shall resume. If the tracked signal has not been received at the end of the memory period, search shall be initiated.

3.4.22.2 Bearing Memory - The equipment shall delay 3 to 8 seconds before searching for a lost bearing signal which had been present previously for at least 15 seconds.

3.4.23 Beacon Identification - In the REC and T/R modes, the equipment shall provide intelligible and unambiguous aural identification of the-selected beacon with a 10 dB signal plus noise-to-noise power ratio within the range of RF signal strength of -90 dBm to 0 dBm. The beacon shall be identifiable by a Morse coded audio tone of 1350 Hz.

3.4.24 Equalizer Pair - The equipment shall not require that equalizer pulse pairs (per MIL-STD-291) be received for proper operation of the identity function.

3.4.25 Outputs

3.4.25.1 RF Input-Output - The RF input and output shall provide for a common connection to a 52 ohm coaxial line. The receiver and transmitter shall not be damaged under any mismatched conditions.

3.5 Detail Requirements

3.5.1 Receiver-Transmitter, TACAN RT-1022/ARN -84(V) The Receiver-Transmitter shall meet the following requirements.

3.5.1.1 Function - The Receiver-Transmitter shall encode and transmit interrogation pulses, reply pulses, squitter pulses, data link pulses and main reference bearing pulses. The Receiver-Transmitter shall receive and process RF TACAN signals and provide digital data for use by the Converter. The Receiver-Transmitter shall also transpond to the interrogation pulses of the similarly equipped aircraft. It shall transmit single pulse replies to as many as five such interrogating aircraft provided their signal strengths are *within* 12 dB of each other at the receiver input terminal.

3.5.1.1.1 Operation- In addition to the operations described in 3.4.1, The Receiver-Transmitter shall perform the

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operations described in Table III. Furthermore, the equipment shall provide and accept the interface signals necessary for an RF data link function in REC (one-way), T/R and A/A (two-way) modes.

3.5.1.2 Form Factor - The form factor of the Receiver-Transmitter shall be in accordance with Figure 2.

3.5.1.3 Weight - The weight of the Receiver-Transmitter shall not exceed 33 pounds.

3.5.1.4 Contents - The Receiver-Transmitter shall contain the following assemblies and subassemblies:

(1) Receiver-Transmitter Chassis

(2) Receiver Module

(3) Decoder/Bearing Computer Module

(4) Range Computer Module

(5) Power Amplifier/Modulator Module

(6) Internal Control Module

(7) Power Supply Module

(8) Circulator Assembly

(9) Self Test Module

(10) Coupler Monitor Module

(11) Frequency Synthesizer Module

(12) Blower Assembly

(13) One Interface Connector To Converter

(14) Two BNC Connectors For Suppression Pulse in/Out

(15) One HN Connector to Antenna

(16) Two Multipin AGE Connectors

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3.5.1.5 Controls - The Receiver-Transmitter shall contain the following controls:

(1) Two screwdriver adjustments marked CTR and WD shall be provided on the front panel of the Receiver-Transmitter for adjusting the course deviation.

(2) A pushbutton switch marked BIT shall be provided to initiate interruptive self test.

(3) A pushbutton switch marked TEMP shall be provided to reset the over temperature indicator.

3.5.1.6 Indicators - The Receiver-Transmitter shall contain the following indicators:

(1) Over Temperature - A manually resettable latching fault type indicator marked TEMP shall be provided to indicate that the internal equipment temperature exceeds component critical levels.

(2) Fault Location - Four magnetic latching FAULT indicators marked RT, ANT, CONT, and CONV shall be provided to indicate a malfunction to the faulty WRA.

3.5.1.7 Electrical Connections - Electrical connections shall be as shown in Figure 3.

3.5.1.8 Modulation Characteristics - Modulation characteristics shall be in accordance with MIL-STD-291 except as modified herein.

3.5.1.8.1 Long Pulse Rise and Fall Time - The time required for the leading edge to rise from 10 to 90 percent of its maximum voltage amplitude shall be 2.5 ± 0.10 microseconds. Correspondingly, the fall time shall be 2.7 ± 0.30 microseconds.

3.5.1.8.2 Pulse Coding (X Mode Channels) - The transmitted interrogating pulse will be coded in pairs with a spacing which shall be 12.0 ± 0.10 microseconds as measured at the 50 percent maximum voltage amplitude points from the leading edge

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of the first pulse to the leading edge of the second pulse. The X-Mode air-to-air interrogating pulse pair spacing shall be 12 ± 0.10 microseconds.

305.108.3 Pulse Coding (Y Mode Channels) - The transmitted interrogating pulses will be coded in pairs with a spacing which shall be 36 ± 0.10 microseconds as measured at the 50 percent maximum voltage amplitude points from the leading edge of the first pulse to the leading edge of the second pulse. The Y Mode air-to-air interrogating pulse pair spacing shall be 24 ± 0.10 microseconds.

3.501.9 Transmitter Power Output

3.5.1.9.1 Transmitter Power in T/R Mode - When the equipment is interrogating a beacon in the T/R mode, the peak power of each pulse transmitted during the search condition shall not be less than 1.5 kw nor more than 3.0 kw on all channels when the transmitter is terminated in a 50 ohm resistive lead impedance. The transmitted peak output power shall not drop below 0.95 kw with a two to one standing wave ratio of any phase angle at the transmitter. The two pulses in an interrogating pulse pair shall not differ in power by more than 0.5 dB when the load impedance is 50 ohms.

3.5.1.9.2 Transmitter Power In A/A Mode - With the equipment operating in the A/A mode at a transmitting rate of 150 PP/S, the requirements of 3.5.1.9.1 shall apply. The transmitter power shall recover to within 2 dB of the maximum peak power within 18 microseconds after any transmission and shall be at least 750 watts when the transmitter rate is 1200 pulses per second.

3.5.1.9.3 Effect of Transmitter Power on Other Functions In the BCN mode, prior to amplitude modulation by the antenna of the composite RF pulse train, the 15 Hz component of any amplitude modulation shall not exceed 1.0 percent. This shall be measured with the transponder operating normally and shall include the effects of droop and recovery time of the main reference group.

3.5.1010 On Frequency and Image Frequency Interference With the equipment tracking on a signal of -80 dBm, the specified performance shall not be affected by the following types and levels of interference at any frequency on or within 3 MHz of the desired or image frequency.

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Type	Image	Level On Freq.
(1) CW (in REC, T/R and A/A modes)		
(1.1) within 400 kHz of tracked signal		
(1.1.1) Bearing	(G/A) -30dBm	-85 dBm
	(A/A) -30dBm	-90dBm
(1.1.2) Beacon Identification	-30dBm	-85dBm
(1.1.3) Distance	-30dBm	-85dBm
(102) 400 kHz to 3 MHz from tracked signal		
(1.2.1) Bearing	-25dBm	-85dBm
(1.2.2) Beacon Identification	-15dBm	-75dBm
(1.2.3) Distance	-15dBm	-75dBm

(2) Pulse pairs of shape conforming to MIL-STD-291 but spaced as follows:

X Channel, Ground-to-Air and Air-to-Air ranging modes: less than 8 microseconds and greater than 16 microseconds.

Y Channel, Ground to Air Mode: less than 26 microseconds and greater than 34 microseconds.

Y Channel, Air-to-Air ranging Mode: less than 20 microseconds and greater than 28 microseconds.

Y Channel, Air-to-Air ranging Mode: IFF (AIMS) reply configuration with Information Pulse (I/P) at 24.65 microseconds.

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The pulse pairs shall be randomly spaced at an average rate of 3,000 pairs per second with a minimum interval between pairs of 60 microseconds at a level of -20 dBm. Also, pulses of 1 microsecond or less paired in accordance with the spacings, rate, and distribution given above at a level of -30 dBm. (In **Receive, Transmit-Receive, and Air-to-Air ranging Modes.)**

(3) Synchronous Signals

(a) Synchronous bearing reference bursts conforming to the main and auxiliary bursts configuration given in MIL-STD-291 and locked to the modulation frequency of the tracked signal but at a phase angle which differs from the reference phase of the tracked signal by 5° to 35° + N 40° (N = any integer). Interfering signal level shall be -80 dBm for the on frequency case and -20 dBm for the image frequency case.

(b) Synchronous DME reply. Early pulses on frequency and identical in level and all other characteristics to the tracked DME reply, except preceding it by an interval of 8 to 16 used in X channel, or 32 to 40 microseconds in Y channel, at a reply efficiency of 50% in lieu of and alternating with the tracked reply (also at 50% reply efficiency). Echo replies identical in level and all other characteristics to the tracked DME reply except following it at an interval greater than 50 microseconds.

3.5.1.11 Outputs

3.5.1.11.1 Antenna Switching - An antenna switching feature shall be incorporated in the equipment for the purpose of of selecting one of two associated antennas. The circuit shall

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accept the first antenna which provides satisfactory signal level. The equipment shall provide either a low impedance (less than 100 ohms) or a high impedance (greater than 10 k ohm) state for the control of the antenna switch. The antenna switch shall not be a part of the equipment specified herein and shall be a type SA-521/A in accordance with Specification MIL-S-25879 or an approved eqivalent. In the absence of a usable signal the antenna selector shall continuously sample the signal delivered from each of the two antennas in an alternating manner such that the antenna transfer control voltage is switched to effect transfer of antennas not less than 3 nor more than 6 seconds after the start of the sampling period for a given antenna.

3.5.1.12 Built-in-Test - The equipment shall contain built-in test features in accordance with AR-10, and as detailed herein. The built-in test features shall include an equipment monitor function and and equipment self-test function.

3.5.1.12.1 Equipment Monitor - Equipment readiness shall be determined by continuous monitoring of critical functional parameters, without interruption of equipment operation. At any time any critical performance parameter falls below the level specified for acceptable operation, a "NO-GO" indication shall be displayed on the control unit. The readiness monitoring function shall, as a minimum, determine the proper operation of the following parameters:

(a) In the T/R Mode the minimum peak RF power level shall be 750 watts. In the A/A and DME-Only Modes, the minimum peak RF power shall be 750 watts for 300 pps and 250 watts for 1650 pps.

(b) VSWR shall be less than 3:1, NO-GO if greater than 6:1. (T/R and A/A).

(c) Bearing measurement (REC, T/R and A/A)

(d) Distance measurement (T/R and A/A)

(e) A/A Transponder (A/A)

(f) Frequency synthesizer shall be in lock.

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(g) Power Supply output voltages

3.5.1.12.2 Self-Test - The equipment shall be capable of performing an interruptive self-test cycle at all positions of the channel selector switch. The self-test cycle shall be initiated. from the Control, the front of the Receiver-Transmitter, and by remote signal, and shall be completed in 22 seconds or less. The self-test function shall provide sufficient testing of the TACAN set to assure a probability of at least 95% of correct detection and isolation of the malfunction to the faulty Weapons Replaceable Assembly (WRA). Upon completion of the self-test cycle, a visual indication of equipment status (GO/NO-GO) shall be provided at the Control. An equipment 'NO-GO" status shall be continuously displayed at the RT front panel until a successful Self-Test cycle is completed.

3.5.1.12.2.1 REC Mode - Initiation of the interruptive self-test cycle shall inject an RF signal with forty percent combined modulation at not more than -70 dBm level to the antenna input terminals. The azimuth indicator shall display a bearing between 0° and 8°. -A clear unambiguous identity tone shall be applied to the pilots headset. If the azimuth indication is not between 0° and 8°, or the identity tone output is not present, "REC NO-GO" shall be indicated.

3.5.1.12.2.2 T/R Mode - Initiation of the interruptive self-test cycle shall inject an RF signal with forty percent combined modulation at not more than -70 dBm level to the antenna input terminals. An indication of "T/R -GO" shall be obtained under all of the following conditions:

(a) Azimuth bearing between 0° and 8°.

(b) Range between 1.6 nm and 2.0 nm

(c) Power output greater than 750 watts

(d) Bearing and Range acquisition times less than 3 seconds

(e) Identity tone output present

(f) Range memory greater than 5 seconds, followed by search

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(g) Bearing memory greater than 2 seconds, followed by search

If any of the above parameters have degraded below the above specified, the indications shall be “T/R NO-GO".

3.5.1.12.2.3 A/A Mode - Initiation of the interruptive self-test shall inject an RF signal at the antenna connector of not more than -70 dBm providing paired pulse interrogations at a rate of 22 pp/s and containing single pulse replies to the TACAN paired pulse interrogations at a reply efficiency of 85 percent. An indication of-”A/A-GO” shall be obtained under the following conditions:

(a) Range between 1.6 nm and 2.0 nm

(b) The TACAN transpond delay at the 50% points on the first received coded pair of RF interrogation pulses and the corresponding transmitted RF reply pulse shall be 62 ± 0.1 microseconds in X mode and 74 ± 0.1 microseconds in Y mode.

(c) Minimum peak RF power shall be 750 watts for 300 pps and 250 watts for 1650 pps.

If any of the above parameters have degraded below the above specified, the indication shall be “A/A-NO-Go”.

3.5.1.12.2.4 Interfering Signals - The self-test shall operate as specified, in the presence of a received TACAN signal of -50 dBm or weaker.

3.5.1.13 Air-To-Air System Delay - The equipment shall effect an indication of 0 ± 0.05 nautical miles in the A/A mode when:

(a) The delay at the 50% leading edge amplitude points of the first transmitted coded pair of RF interrogation pulses and the corresponding received RF reply pulse is 62 ± 0.1 microseconds in X mode and 74 ± 0.1 microseconds in Y mode.

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(b) The delay at the 50% leading edge amplitude point of the first received coded pair of RF interrogation pulses and the corresponding transmitted RF reply pulse shall be 62 ± 0.1 microseconds in X mode and 74 ± 0.1 microseconds in Y mode.

3.5.1.14 Airborne Beacon - In addition to the requirements specified elsewhere herein, this equipment shall be capable of determining and indicating the bearing between carrying aircraft and other complementary aircraft equipped with the AN/ARN-90; or the AN/ARN-84 and an antenna producing a 15 Hz modulated rotating pattern with associated interfacing circuitry. The interfacing circuitry shall not be part of the AN/ARN-84 and shall meet the following requirements:

(a) Receive power and aircraft magnetic heading information from the AN/ARN-84.

(b) Provide circuitry necessary to produce the 15 Hz antenna pattern rotation.

(c) Provide North Reference triggers to the AN/ARN-84 modulator.

3.5.1.14.1 Transmitting Requirements - The equipment when operating with an antenna producing a 15 Hz modulated rotating pattern and associated interfacing circuitry shall transmit a 15 Hz ± 0.2% modulated pulse train consisting of: (1) Single pulses or substitute squitter pulses at a rate of 1350 ± 300 pulses per second (distribution shall be generally random except that the minimum interval between transmissions shall be 60 us), and (2) reference bursts consisting of 10 pulses spaced 30 ± .5 microseconds apart. With the equipment operating in the BCN mode on a given channel, it shall be possible to transpond effectively to each of five interrogating aircraft when their equipments are set 63 channels from the given channel and when their signal strength at the antenna input connector of the equipment specified herein are within 12 to 15 dB of each other when the level of the weakest signal is any value from -90 dBm to -l2 dBm. The equipment specified herein shall interrogate and lock-on to one of the five, but it shall not be required to determine the one to which it is locked-on. The equipment shall not be damaged when subjected to any number of interrogations.

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3.5.1.14.1.1 Power - Transmitter power in the BCN mode shall not be less than 600 watts at 1650 pulses per second, including reference bursts, when the load impedance is 50 ± 5 ohms resistive l

3.5.1.14.1.2 Pulse Droop - Pulse droop shall not exceed 3 dB.

3.5.1.15 Inverse TACAN - In addition to the requirements specified elsewhere herein, the equipment when used with an antenna producing a 15 Hz modulated rotating pattern with associated interfacing circuitry, shall measure distance and bearing (referred to magnetic north) to complementary beacons (airborne or surface) transmitting only squitter pulses and/or DME replies.

3.5.1.15.1 Accuracy

3.5.1.15.1.1 Distance - The distance measurement accuracies shall be as shown in Table II.

3.5.1.15.1.2 Bearing - The accuracy of the Inverse TACAN bearing measurements shall be as follows:

Digital Accuracy	Analog Accuracy	Signal Strength
± 2.0°	± 2.5°	-82 dBm to 0 dBm
± 2.5°	±3.0°	Below -82 dBm

This requirement shall be applicable over a 7% to 40% range of modulation with and without auxiliary reference groups.

3.5.1.15.2 Reply and Squitter Pulse Distribution - The Inverse TACAN accuracies shall be maintained for beacons transmitting only DME replies and/or squitter pulses varying between 700 and 2700 pulses per second.

3.5.1.16 DME Only Beacon - In addition to the above, the equipment shall be capable of transmitting single pulse replies or substitute squitter pulses at a rate of 1350 ± 300 pulses per second. Distribution shall be generally random except that the minimum interval between transmissions shall be 60 microseconds. This requirement applies when using a standard TACAN omni-directional airborne antenna.

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3.5.1.16.1 Power - The transmitter power shall not be less than 600 watts at a pulse repetition frequency of 1650 pulses per second.

3.5.1.16.2 Pulse Droop - Pulse droop shall not exceed 3 dB.

3.5.1.17 Data Link - When triggered by external triggering pulses of 5.5.1.17.2, the equipment shall be capable of transmitting pulse groups of 3 TACAN pulses spaced 18 or more microseconds apart. These groups may be assembled into words or frames consisting of up to seven groups spaced 205 to 452 used apart. These frames shall be evenly spaced and the frame rate must be consistent with the transmitter duty cycle and A/A range and bearing requirements; i.e., for a pulse rate of 1200 pps or less, the minimum peak power is 750 watts and for a pulse rate of 1650 pps, the minimum peak power is 600 watts. Pulse droop measured from the first pulse in a frame to the last pulse in the same frame, shall not exceed 3 dB. Incoming detected TACAN pulses shall be available for processing by a data link adapter.

3.5.1.17.1 External Transmitter Enable - The transmitter circuits shall accept a digital signal, which in the logical "O" state shall interrupt internal triggers to the transmitter and permit external triggering of the transmitter. This signal shall bracket external transmitter triggers.

3.5.1.17.2 External Transmitter Trigger - The transmitter section shall accept external triggers of duration between 1 and 5 microseconds, and shall generate properly shaped TACAN pulses and transmit them with a delay of 16 ± 0.5 microseconds. Minimum spacing of the triggers shall be 18 microseconds.

3.5.1.17.3 Receiver Video Output - A receiver video output shall be provided having positive linear video pulses of **6.5 ± 1.5** volts amplitude when measured with no modulation.

3.5.1.18 Audio Modulation Frequency Selectivity - The equipment shall minimize the susceptibility of the bearing measurement to the effects of extraneous modulation such as that introduced by rotors, propellers or signal strength fluctuations. As a minimum, signal strength fluctuations of ±50% at a rate of twice per second shall not degrade the bearing accuracy.

3.5.2 Receiver-Transmitter, TACAN RT-1127/ARN-84(V) The Receiver-Transmitter shall meet the-foll**owing** requirements,

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3.5.2.1 Function - The Receiver-Transmitter shall encode and transmit interrogation and reply pulses. The Receiver-Transmitter shall receive and process RF TACAN signals and provide digital data outputs. The Receiver-Transmitter shall also transpond to the interrogation pulses of the similarly equipped aircraft. It shall transmit single pulse replies to as many as five such interrogating aircraft provided their signal strengths are within 12 dB of each other at the receiver input terminal.

3.5.2.2 Form Factor - The form factor of the Receiver-Transmitter shall be in accordance with Figure 4.

3.5.2.3 Weight - The weight of the Receiver-Transmitter shall not exceed 27 pounds.

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3.5.2.4 Contents - The Receiver-Transmitter shall contain the following assemblies and subassemblies:

(1) Receiver-Transmitter Chassis

(2) Receiver Module

(3) Power Amplifier/Modulator Module

(4) Power Supply Module

(5) Circulator Assembly

(6) Coupler Monitor Module

(7) Frequency Synthesizer Module

(8) Data Processor Module

(9) Internal Blower Assembly

(lo) One Interface Connector to Converter or Mount

(11) Two BNC Connectors for Suppression Pulse In/Out

(12) Two HN Connectors to Antenna or Converter

(13) One Multipin AGE Connector

3.5.2.5 Controls - The Receiver-Transmitter shall have two screwdriver adjustments marked CTR and WD on the front panel of the Receiver Transmitter for adjusting the course deviation.

3.5.2.6 Electrical Connections - Electrical connections shall be as shown in Figure 5.

3.5.2.7 Modulation Characteristics - Modulation characteristics shall be in accordance with MIL-STD-291.

3.5.2.8 Transmitter Power Output

3.5.2.8.1 Transmitter Power in T/R Mode - When the equipment is interrogating a beacon in the T/R mode, the peak power of each pulse transmitted during the search condition shall not be less than 0.75 kw not more than 3.0 kw on all channels when the transmitter is terminated in 50 ohm resistive load impedance. The transmitted peak output power shall not drop below 0.65 kw with a two-to-one standing wave ratio of any “phase angle at the transmitter. The two pulses in an interrogating pulse pair shall

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not differ in power by more than 0.5 dB when the load impedance is 50 ohms.

3.5.2.8.2 Transmitter Power in A\A Mode - With the equipment operating in the A/A mode at a transmitting rate of 150 pps, the requirements of 3.5.2.8.1 shall apply. The transmitter power shall recover to within 2 dB of the maximum peak power within 200 microseconds after any transmission and shall be at least 500 watts when the transmitter rate is 1050 pulses per second.

3.5.2.9 Electromagnetic Interference Control - The equipment shall meet the following requirements of MIL-STD-461: CE01, CE02, CE03, CE04, CE06, RE02, CS01, CS02, CS06, and RS02. Requirement RS03 as modified by paragraph 3.5.2.9.3 shall also be applicable. Requirements CS03, CS04 and CS05 are not applicable. The requirements of paragraphs 3.4.13.1, 3.4.13.2, 3.5.2.9.1, and 3.5.2.9.2 shall apply.

3.5.2.9.1 On Frequency and Image Frequency - With the equipment tracking on a signal of -80 dBm, the specified performance shall not be affected by the following types and levels of interference at any frequency on or within 3 megahertz of the desired or image frequency.

(1) CW (in REC, T\R and A\A modes)

(1.1) on or within 3 MHz of tracked signal

		Image	On Freq.
(1.1.1)	Bearing	-50 dBm	-110 dBm
(1.1.2)	Beacon Identification	-35 dBm	- 95 dBm
(1.1.3)	Distance	-35 dBm	- 95 dBm

3.5.2.9.2 Off Frequency Interference - The requirements of 3.4.13.2 shall apply, except that the CW signal introduced to the antenna while the equipment is tracking a TACAN signal at a level of -80 dBm, shall have an amplitude of -65 dBm for frequencies between 3 MHz and 10 MHz of the desired signal.

3.5.2.9.3 Radio Frequency Radiated Susceptibility - No degradation of performance shall occur when the equipment is subjected to a radio frequency field of one volt per meter over the frequency range of 14 kHz to 10 GHz according to the limits for RS03.

3.5.2.10 Antenna Switching Output - An antenna switching circuit shall be incorporated in the unit for the purpose of selecting one of two associated antennas. The circuit shall stop

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the external antenna switch on the antenna that provides a minimum satisfactory signal level for lock-on. In the absence of a usable signal, the antenna switch shall switch the receiver between the two antenna ports at a rate of 3 to 6 seconds. An internal solid state antenna switch will be provided as part of the equipment. By not grounding the antenna switch enable lead, the switching circuit shall connect the unit to antenna number one connection.

3.5.2.11 Built-In-Test - The equipment shall contain built-in-test features that shall provide a positive visual indication on the airplane cockpit displays if the equipment is operating satisfactorily. The entire BIT procedure can be performed from the cockpit without access to equipment not available in the cockpit. BIT shall provide a confidence level of approximately 90% of detecting a fault and isolating it to the WRA. The BIT procedure shall include an interrupting self test cycle which is initiated by pressing a button on the Control Unit. When the cycle is initiated, the following functions shall be checked and the results displayed on the appropriate indicator on the airplane cockpit.

(a) The Control Unit shall be checked by inserting a test word in the Control Unit and checking to see that it is properly received by the R\T Unit.

(b) The R/T Unit shall be checked by measuring various critical signals in the R\T circuits for an in-tolerance condition while an internally generated RF TACAN type signal is present at the receiver input.

The TACAN System Status shall be checked by monitoring the above functions. If all are in GO status, the System Status shall be displayed as a range readout of 0.0 (+0.5, -0.0) nautical mile and a bearing indication of 180.0 ± 2.5 degrees.

3.5.2.12 Identity Tone Output - The identity tone output shall be a 1350 Hz signal. The maximum output shall be not less than 75 mw into a 150 ohm resistive load. The audio output shall not vary more than 3 dB over an RF input signal range of -90 dBm to 0 dBm. Accidental shorting or opening of the output circuit external to the equipment shall not cause damage to the equipment or render other functions of the equipment inoperative.

3.5.2.13 Air-to-Air System Delay - The equipment shall effect an indication to 0 ± 0.5 nautical mile in the A\A Mode when:

(a) The delay at the 50% leading edge amplitude points of the first transmitted coded pair of RF interrogation pulses and

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the corresponding received RF reply pulse is 62 ± 0.1 microseconds in X mode and 74 ± 0.1 microseconds in Y mode.

(b) The delay at the 50% leading edge amplitude point of the first received coded pair of RF interrogation pulses and the corresponding transmitted RF reply pulses shall be 62 ± 0.5 microseconds in X mode and 74 ± 0.5 microseconds in Y mode.

3.5.3 Control, Navigation Set C-8734/ARN-84(V) - The Control shall meet the applicable requirements of Specification MIL-C-6781 for Type 1 control panel. The lighting of the plastic panel shall conform to MIL-P-7788, Class 1-R, Type IV with a lamp voltage of 28 volts.

3.5.3.1 Function - The function of the Control shall be to control the operation of the TACAN Set, to select the operating channel, and to indicate the status of the Set.

3.5.3.2 Form Factor - The form factor of the Control shall be in accordance with Figure 6.

3.5.3.3 Weight - The weight of the Control shall not exceed 2.0 pounds.

3.5.3.4 Contents - The Control shall contain two circuit card assemblies and one interface connector.

3.5.3.5 Indicators - The Control shall have the following indicators:

(1) STATUS GO - A green lamp to indicate that self test was successful.

(2) STATUS NO-GO - An amber lamp to indicate that self test detects a fault in the TACAN.

3.5.3.6 Controls - The Control shall have the following controls:

(1) CHAN - The channel selector shall consist of rotary switches. The “units” switch shall contain 10 positions and shall effect a digital readout of zero to nine respectively as the switch is moved clockwise through its range. The “tens/hundreds” switch shall contain 13 positions and shall effect a digital readout of one to twelve as the switch is moved clockwise from its second position

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to its thirteenth position respectively. There shall be no readout in the most counterclockwise position. The digits selected by the switches shall be displayed in such a manner as to optimize their readability from either side.

(2) MODE - A toggle switch to select NORMAL or INVERSE.

(3) X/Y - A toggle switch to select either mode X or mode Y channelization.

(4) VOL - A potentiometer assembly which shall provide continuous variation of the audio output level, approximately linearly in dB per degree of control rotation, over a range of attenuation required to meet the system output requirements. A clockwise rotation shall effect an increase in output level.

(5) Function - A five position rotary switch that when

(a) OFF - Removes primary power from the equipment.

(b) REC- Operates the equipment in a receive only mode to measure bearing and receive station identity.

(c) T/R - Transmits interrogations and receive replies to measure bearing, slant range, and receive station identity.

(d) A/A - Transmits and receives interrogations or replies to measure slant range to or from a complementary aircraft and measures relative bearing to a properly equipped cooperating aircraft.

(e) BCN- Operates as an airborne beacon to cooperating aircraft.

(6) BIT - A pushbutton switch to initiate self test.

3.5.3.7 Electrical Connections - Electrical connections to the Control shall be as shown in Figure 3.

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3.5.4 Control, Navigation Set C-9054\ARN-84(V) - The Control shall meet the applicable requirements of Specification MIL-C-6781 for Type 1 control panel. The lighting of the plastic panel shall conform to MIL-P-7788, Class l-R, Type IV with a lamp voltage of 5 volts. The Control shall meet the requirements of 3.5.3.1 through 3.5.3.7 inclusively.

3.5.5 Control, Navigation Set C-9475/ARN-84(V)- The Control shall meet e requirements of MIL-C-6781 for Type 1 control panel. The lighting of the plastic panel shall conform to MIL-P-7788, Class l-W, Type IV with a lamp voltage of 5 volts

3.5.5.1 Function - The Control shall perform the same functions as shown in 3.5.3.1.

3.5.5.2 Form Factor - The form factor of the Control shall be in accordance with Figure 7.

3.5.5.3 Weight - The weight of the Control shall not exceed 2.0 pounds.

3.5.5.4 Contents - The Control shall contain one terminal board assembly and one interface connector.

3.5.5.5 Indicators - The Control shall not contain any indicators.

3.5.5.6 Controls - The Control shall have the following controls:

(1) CHAN - The channel selector shall perform the same functions as shown in 3.5.3.6(l).

(2) X/Y - A two position rotary switch to select either mode X or mode Y channelization.

(3) VOL - The volume control shall perform the same functions as shown in 3.5.3.6(4) .

(4) Function - A four position rotary switch that performs the same functions as shown in 3.5.3.6(5) (a) through 3.5.3.6(5)(d) inclusively.

(5) TEST - A pushbutton switch to initiate

3.5.5.7 Electrical Connections - Electrical connections to the Control shall be as shown in Figures 13, 17 and 19.

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3.5.6 Control, Navigation Set C-9483/ARN-84(V) The Control shall meet the applicable requirements of Specification MIL-C-6781 for Type 1 control panel. The lighting of the plastic panel shall conform to MIL-P-7788, Class 1-R, Type IV with a lamp voltage of 5 volts. The Control shall meet the requirements of 3.5.5.1 through 3.5.5.7 inclusively.

3.5.7 Converter, Signal Data CV-2837/ARN-84(V)

3.5.7.1 Function - The functions of the Signal Data Converter shall be to:

(1) Transform digital inputs from the Receiver-Transmitter to synchro format to operate the aircraft instruments.

(2) Furnish signals for the AN/ASW-27 Data Link Set; specifically range and bearing, in analog form, station number in natural binary form, and discretes.

(3) Provides the interface to transform parallel inputs from the Receiver-Transmitter to serial data outputs for use by an external computer.

3.5.7.2 Form Factor - The form factor of the Signal Data Converter shall be in accordance with Figure 2.

3.5.7.3 Weight - The weight of the Signal Data Converter shall not exceed 11 pounds.

3.5.7.4 Contents - The Signal Data Converter shall contain the following assemblies and subassemblies:

(1) Buffer Module

(2) Interface Module

(3) Range Coupler Module

(4) Bearing Coupler Module

3.5.7.5 Controls - The Signal Data Converter shall not contain any controls or Indicators other than an elapsed time indicator.

3.5.7.6 Electrical Connections - Electrical connections shall he in accordance with Figure 3. -

3.5.7.7 Analog Bearing Outputs - The equipment shall provide analog bearing outputs for display on any combinations of three ID-663/U, ID-1013A, or AQU-4A Indicators. A bearing output presentation accuracy degradation of ± 0.5 degree, other than

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reading error, is “permissible when the equipment is loaded with three ID-663 (other than ID-663C) indicator instruments in parallel. The range of the output is 0 to 360 degrees and the scale factor shall be 1° /l°. The analog bearing output shall be in synchro format in accordance with MIL-S-20708.

3.5.7.8 Analog Range Output - The equipment shall provide analog range outputs for display on any combination of three ID-663/U, ID-1013A, or AQU-4A Indicators. The analog range output shall consist of three synchro outputs in accordance with MIL-S-20708; one each synchro output for units, tens, and hundreds.

3.5.7*9 Course Deviation Output - The equipment shall provide course deviation outputs from a manually selected course to or from a beacon for display on from one to three ID-387/ARN Course Indicators, and shall be a current type output. Course deviation sensitivity and centering shall be adjustable from the front panel of the Receiver-Transmitter so that deviations from 2.5° to 15° will provide 450 microampere into a constant load impedance of 333 ohms.

3.5.7.10 To-From Flag - The equipment shall provide outputs to operate the To-From Flag in the ID-387/ARN Course Indicator.

3.5.7.11 Bearing Flag Alarm - The flag alarm of the ID-387/ARN shall be actuated and the flag shall disappear when reliable signals above minimum usable are received. Upon loss of signal, the flag shall reappear at expiration of the memory period. The flag shall be in view in the absence of 15 Hz signals. Flag operation is also required when the percentage of modulation of the 15 Hz and 135 Hz signal is below the point where adequate phase stability between the two signals can be maintained. Flag operation shall be compatible with the bearing accuracy requirements of the equipment.

3.5.7.12 Distance Flag Alarm - The equipment shall be capable of operating the flag alarm of associated indicators to show either presence or absence of an adequate signal except during the memory period.

3.5.7.13 Identity Tone Output - The identity tone output shall be a 1350 Hz signal. The output shall be greater than 100 milliwatts when loaded with a 600 ohm resistive load. The maximum output impedance shall be 600 ohms. The audio output shall not vary more than 3 dB over an RF input signal range of -90 dBm to 0 dBm. Accidental shorting or opening of the output circuit external to the equipment shall not cause damage to the equipment or render other functions of the equipment inoperative.

3.5.7.14 Outputs for Digital Computer - The Converter shall provide station number, distance and bearing outputs in

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digital, serial form to interface with sundry computers used in Navy aircraft. The interface, computer to TACAN, is composed of the four signals; READ, ADDRESS, DATA OUT, and SHIFT CLOCKS. The READ line is a single wire discrete. The ADDRESS signal is composed of two single wire discretes. The SHIFT CLOCKS are received by the TACAN as a differential output from the computer. The DATA OUT shall be received by a double ended receiver.

The READ line shall be simplexed to the TACAN. The ADDRESS lines shall be valid a minimum of 0.5 microseconds prior to the raising of the READ line. The READ and ADDRESS lines shall be valid a minimum of 0.5 microsecond prior to the transmission of SHIFT CLOCKS . A maximum of 200 nanoseconds after the READ line is brought up, bit #l of data shall be available on the DATA OUT lines. The TACAN output register shall then be shifted *once* on each of the 20 positive (voltage) transitions of the SHIFT CLOCK provided by the Computer.

The data will be received from TACAN serially at a rate of 5 kHz to 1 MHz. At a minimum of one microsecond after the 20th SHIFT CLOCK pulse from the computer, the READ line will be dropped. The raising of the READ line shall then be inhibited for a minimum of 10 microseconds after it has been dropped. Minimum and Maximum times shall be measured between the 90% point of the previous transition and the 10% point of the subsequent transition.

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3.5.7.1401 Data Word Format - The data will be transferred to the computer in the format as follows:

Bit	Designation	Value		
		Range	Bearing	Station Number
1	Control	0	0	0
2	Validity	1 = Valid	1 = Valid	1 = Valid
3	Data LSB	1 = 0.025	1 = 0.25	LSB = 1
4				2
5				Binary 4
6				Coded 8
7			Straight	Decimal 10
8		Straight	Binary	20
9		Binary	(Degrees)	40
10		(Nautical		MSB=80
11		Miles)		0
12				0
13			1 = 256.0	0
14			0	0
15			0	0
16		1 = 204.8	0	0
17	Data MSB	0	0	0
18	Sign	0 = Positive	0 = Positive	0 = X Station
				1 = Y Station
19	Spare	0	0	0
20	Parity	ODD	ODD	ODD

The transmission sequence shall be such that the control bit is transmitted first. All data shall be least significant bit justified and logical zeros shall be inserted into the remaining most significant data bits beyond the resolution requirements. The binary bearing word for zero degrees shall be equivalent to the binary word for 360.00 degrees and shall consist of all logical zeros. The parity bit is assigned a value such that the total number of ones in the 20-bit word is odd.

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3.5.7 .14.2 Computer TACAN Data Channel

Signal Name	Signal Type
DATA OUT - True	Differential Non-return-to zero (NRZ)
DATA OUT - One's Complement	Differential Non-return-to-zero (NRZ)
SHIFT CLOCK - True	Differential 5 kHz to 1 MHz, 50±10% Duty Cycle
SHIFT CLOCK - One’s Compl l	Differential 5 kHz to 1 MHz, 50±10% Duty Cycle
READ	Discrete “1” Read
ADDRESS - Line 1 ADDRESS - Line 2	Signal Code Line 2 / Line 1 / Bearing 0 / 0 0 / 1 / Range 1 / 0 / Station Number
SIGNAL GROUND	All signal grounds are common and tied to chassis ground

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3.5,7.14.3 TACAN Output Differential Transmitter - The TACAN DATA OUT transmitter shall be capable of driving a line which has up to 1,000 picofarads capacitance, terminated in a resistive load of 2,500 ohms minimum, 10,000 ohms maximum, as measured between the true and its complement. Protection shall be provided such that a short circuit of the output terminals to ground does not result in damage to other circuit components or affect any other channel. When power is removed from a line driver, its output impedance shall be greater than 10,000 ohms.

Logical O	+5.0 ± 1.0 Volts with the transmitter sourcing 5.0 milliamperes maximum.
Logical 1	0 ± 0.5 Volts with the transmitter sinking 5.0 milliamperes maximum.
Rise/Fall Time	The rise/fall time shall be a maximum of 10% of the clock period provided by the computer.

3.5.7.14.4 TACAN Input Differential Receiver - The TACAN SHIFT CLOCK receiver shall be of the double ended, differential amplifier type, Signal detection with reduced noise discrimination shall be effected if one input becomes open circuited. If both inputs become open circuited, (source impedance exceeds 10,000 ohms) the receiver shall operate as though a logical zero were present at its input. The receiver shall be capable of receiving signals having the following characteristics:

Logical 0	+5.0 ± 1.0 Volts with the receiver sinking 2.0 milliamperes maximum.
Logical 1	0 ± 0.5 Volts with the receiver sourcing 2.0 milliamperes maximum.
TACAN Input Capacitance	500 picofarads maximum.
Common Mode Voltage	±5 Volts maximum, with respect to ground.

3.5.7.14.5 TACAN Input Discrete Receiver - The TACAN READ and ADDRESS discrete inputs shall single ended receivers. If a discrete input becomes open circuited, the receiver shall operate as though a logical O were present at its inputs. The receiver shall be capable of receiving signals, referenced to signal ground, with the following characteristics:

Logical O	+5.0 ± 1.0 Volts with the receiver sinking 2.0 milliamperes maximum.
Logical 1	0 ± 0.5 Volts with the receiver sourcing 2.0 milliamperes maximum.

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3.5.7.15 AN/ASW-27 A/A Data Link Interface - The TACAN Equipment shall provide the following output signals for direct input to the AN/ASW-27 A/B or approved equivalent Data Link subsystem.

3.5.7.15.1 Quantized Analog Range - Measured TACAN range shall be represented in a quantized analog form consisting of positive pulses whose duration is directly proportional to measured range. Range pulse characteristics, when loaded by 91 ohms and 500 pf to ground, shall be as follows:

Logical 1	5.0 Volts dc ± 1.0 Volts dc
Logical 0	0.0 Volts dc ± 0.5 Volts dc
Rise Time	10% to 90% amplitude, 0.5 microsecond maximum
Fall Time	90% to 10% of amplitude, 0.5 microsecond maximum
Pulse duration when measured at the pulse 50% point:	
	0 nautical miles - 9.9 microsecond ± 1.0 microsec.
	300 nautical miles - 3717 microsecond ± 3.5 microsec.
Repetition rate:	140 pulses per second nominal
	20 pulses per second minimum

3.5.7.15.2 Quantized Analog Bearing - Measured TACAN bearing shall be provided in a quantized se analog form consisting of a reference rectangular wave and a variable -rectangular wave whose phase difference represents the measured TACAN bearing. Bearing rectangular wave characteristics when loaded by 91 ohms and 500 pf to ground shall be as follows:

Logical 1	5.0 Volts dc ± 1.0 Volts dc
Logical 0	0.0 Volts dc ± 0.5 Volts dc
Rise Time	10% to 90% of amplitude - 1.0 microsecond maximum
Fall Time	90% to 10% of amplitude - 1.0 microsecond maximum

Repetition frequency (reference and variable waves)-15±l% per second.

The range of phase difference between the positive edge of the reference pulse and the positive edge of the variable pulse shall be 0.0 ± 0.3 degrees to 359.8 ± 0.3 degrees.

3.5.7.15.3 TACAN Bearing Clock - A 54 kHz clock output shall be provided to determine TACAN bearing phase delay". Zero

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clocks shall equal zero degrees and 3599 clocks shall equal 359.9 degrees. The clock shall effectively divide the total rectangular wave period into 3600 equal parts. Clock pulse characteristics when loaded with 91 ohms and 500 pf to ground shall be as follows:

Logical 1	5.0 Volts dc ± 1.0 Volts dc
Logical 0	0.0 Volts dc ± 0.5 Volts dc
Rise Time	10% to 90% of amplitude - 2.0 microseconds maximum
Fall Time	90% to 10% of amplitude - 2.0 microseconds maximum

3.5.7.15.4 TACAN Bearing Validity - A signal shall be provided to indicate when TACAN bearing is valid. Bearing validity signal characteristics when loaded with 91 ohms and 500 pf to ground shall be as follows:

Bearing Valid	12.0 to 30.0 Volts dc continuous
Bearing Not Valid	0.0 Volts dc ± 0.5 Volts dc continuous

3.5.7.15.5 TACAN Range Validity - A signal shall be provided to indicate when TACAN range is valid. Range validity signal characteristics shall be as follows:

Range Valid	12.0 to 30.0 Volts dc continuous
Range Not Valid	0,0 Volts dc ± 0.5 Volts dc continuous

3.5.7.15.6 TACAN X/Y Select - A signal shall be provided to indicate the selected channel type l The X/Y select signal characteristics shall be as follows:

"X" Channel Selected	- 12 to 30 Volts dc continuous
"Y" Channel Selected	- 0.0 ± 0.5 Volts dc continuous

The source impedance which the TACAN X/Y Select signal generator shall exhibit are as follows:

"x" Channel Selected	- greater than 1000 ohms
"Y" Channel Selected	- less than 100 ohms

3.5.7.15.7 TACAN Station Identity - TACAN channels 1-126 shall be Provided to Data Link encoded on 7 lines in natural binary form. Provisions shall be included to set the binary TACAN station lines to binary zero when the air-to-air mode is selected. TACAN station signal characteristics shall be as follows:

Logical 1	0.0 Volts dc ± 0.5 Volts dc
Logical 0	12 to 30 volts dc

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The source impedance characteristics which the TACAN station identity signal generator shall exhibit are as follows:

Logical 1	less than 100 ohms
Logical 0	greater than 1000 ohms

3.5.7.16 Odd-Even Channel Output - A discrete output signal shall be provided to indicate that an odd or even channel has been selected. - A +28 ± 4 Volts dc (open circuit) signal, with a source impedance greater than 75K ohms, shall indicate that an even numbered channel has been selected. A 0.0 ± 1.0 Volts, dc signal, with a source impedance less than 100 ohms, shall indicate that an odd numbered channel has been selected.

3.5.7.17 Air-to-Air Control Output - A discrete output signal of +28 ± 4 Volts dc (open circuit) amplitude, with a source impedance greater than 75K ohms, shall indicate that the Air-to-Air mode of operation has been selected.

3.5.7.18 Referenced Decode Output - A pulsed output signal shall be provided to indicate that the equipment has decoded a main reference burst.

3.5.7.19 Power Control Signal - A discrete output signal of +25 ± 5 Volts dc amplitude, shall he provided in the Receive, Transmit-Receive, and Air-to-Air Modes and associated equipment for the purpose of external power control.

3.5.8 Converter, Signal Data CV-3135/ARN-84(V)

3.5.8.1 Function - The Signal Data Converter shall provide the conversion of data from the Receiver-Transmitter RT-1127/ARN-84(V) to flight instruments for presentation of slant range and/or bearing of an aircraft to a ground navigation beacon or another aircraft.

3.5.8.2 Form Factor - The form factor of the Signal Data Converter shall be in accordance with Figure 8.

3.5.8.3 Weight - The weight of the Signal Data Converter shall not exceed 9.0 pounds.

3.5.8.4 Contents - The Signal Data Converter shall contain the following assemblies and subassemblies:

(1) Buffer Module

(2) Interface Module

(3) Range Coupler

(4) Bearing Coupler

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(5) Chassis

(6) One Interface Connector to Mount

3.5.8.5 Indicators - The Signal Data Converter shall contain two magnetic latching FAULT indicators marked RT and CV in addition to the elapsed time indicator.

3.5.8.6 Controls - The Signal Data Converter shall not contain any controls.

3.5.8.7 Electrical Connections - Electrical connections shall be in accordance with Figure 9.

3.5.8.8 Outputs - The outputs of the Signal Data Converter shall meet all the requirements of paragraphs 3.5.7.7 through 3.5.7.12 inclusively.

3.5.9 Converter, Signal Data CV-3134/ARN-84(V)

3.5.9.1 Function - The Signal Data Converter shall perform the same functions as shown in 3.5.8.1.

3.5.9.2 Form Factor - The form factor of the Signal Data Converter shall be in accordance with Figure 10.

3.5.9.3 Weight - The weight of the Signal Data Converter-shall not exceed 9.0 pounds.

3.5.9.4 Contents - The Signal Data Converter shall contain the following assemblies and subassemblies:

(1) Buffer Module

(2) Inferface Module

(3) Range Coupler

(4) Bearing Coupler

(5) Chassis

(6) One Interface Connector to Mount

3.5.9.5 Indicators - The Signal Data Converter shall contain two magnetic latching FAULT indicators marked RT and CV in addition to the elapsed time indicator.

3.5.9.6 Controls - The Signal Data Converter shall not contain any controls.

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3.5.9.7 Electrical Connections Electrical connections shall be in accordance with Figure 11.

3.5.9.8 Outputs - The outputs of the Signal Data Converter shall meet a e requirements of paragraphs 3.5.7.7 through 3.5.7.12 inclusively.

3.5.10 Converter, Signal Data CV-3136\ARN-84(V)

3.5.10.1 Function - The Signal Data Converter shall perform the same functions as shown in 3.5.8.1.

3.5.10.2 Form Factor - The form factor of the Signal Data Converter shall be in accordance with Figure 12.

3.5.10.3 Weight - The weight of the Signal Data Converter shall not exceed 14.0 pounds.

3.5.10.4 Contents - The Signal Data Converter shall contain the following assemblies and subassemblies:

(1) Buffer Module

(2) Interface Module

(3) Range Coupler

(4) Bearing Coupler

(5) Chassis

(6) Two Interface Connectors to Aircraft Instruments and Power

(7) Two BNC Connectors for Suppression Pulse In/Out

(8) One HN Connector to Antenna

(9) One Interface Connector to RT

3.5.10.5 Indicators - The Signal Data Converter shall contain two magnetic latching FAULT indicators marked RT and CV in addition to the elapsed time indicator.

3.5.10.6 Controls - The Signal Data Converter shall not contain any controls.

3.5.10.7 Electrical Connections -Electrical connections shall be in accordance with Figure 13.

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3.5.10.8 Outputs - The outputs of the Signal Data Converter shall meet all the requirements of paragraphs 3.5.7.7 through 3.5.7.12 inclusively.

3.5.11 Base, Shockmount, Electrical Equipment MT-4354/ARN-84(V) - The Base shall meet the applicable requirements of MIL-M-81288(WP).

3.5.11.1 Function - The function of the Base shall be to provide mounting of the Receiver-Transmitter and the Signal Data Converter and provide for electrical connections to aircraft wiring.

3.5.11.2 Form Factor - The form factor of the Base shall be in accordance with Figure 14.

3.5.11.3 Weight - The weight of the Base shall not exceed 8.0 pounds.

3.5.11.4 Contents - The Base shall contain the following assemblies:

(1) Mounting base

(2) Foam vibration isolator

(3) Six electrical connectors

3.5.11.5 Controls and Indicators - The Base shall not contain any controls or indicators.

3.5.11.6 Electrical Connections - The electrical connections shall be in accordance with Figure 3.

3.5.12 Base, Hardmount, Electrical Equipment MT-4443/ARN-84(V)- The Base shall meet the applicable requirements of MIL-C-172.

3.5.12.1 Function - The Base shall provide suitable mounting provisions for the Receiver-Transmitter and Converter, Signal Data and shall provide for electrical connections.

3.5.12.2 Form Factor - The Base shall conform to the dimensional requirements for the Receiver-Transmitter and the Converter, Signal Data. See Figure 15.

3.5.12.3 Weight - The weight of the Base shall not exceed 6 pounds.

3.5.12.4 Contents - The Base shall contain the following assemblies:

(1) Mounting base

(2) Six electrical connectors

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3.5.12.5 Controls and Indicators - The Base shall not contain any controls or indicators.

3.5.12.6 Electrical Connections - The electrical connections shall be in accordance with Figure 3.

3.5.13 Base, Shockmount, Electrical Equipment MT-4616\ARN-84(V) - The Base shall meet the applicable requirements of MIL-C-172.

3.5.13.1 Function - The Base shall provide suitable mounting provisions for the Receiver-Transmitter RT-1127/ARN-84(V) and Signal Data Converter CV-3135/ARN-84(V), and shall provide for electrical connections.

3.5.13.2 Form Factor - The form factor of the Base shall be in accordance with Figure 16.

3.5.13.3 Weight - The weight of the Base shall not exceed 8.0 pounds.

3.5.13.4 Contents - The Base shall contain the following assemblies:

(1) Mounting Base

(2) Four Shock Isolators

(3) Six electrical connectors

3.5.13.5 Controls and Indicators - The Base shall not contain any controls or indicators.

3.5.13.6 Electrical Connections - The electrical connections shall be an accordance with Figure 17.

3.5.14 Base, Shockmount, Electrical Equipment MT-4615\ARN-84(V) - The Base shall meet the applicable requirements of MIL-C-l72.

3.5.14.1 Function - The Base shall provide suitable mounting provisions for the Receiver-Transmitter RT-1127\ARN-84(V) and Signal Data Converter CV-3134\ARN-84(V), and shall provide for electrical connections.

3.5.14.2 Form Factor - The form factor of the Base shall be in accordance with Figure 18.

3.5.14.3 Weight - The weight-of the Base shall not exceed 8.0 pounds.

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3.5.14.4 Contents - The Base shall contain the following assemblies:

(1) Mounting Base

(2) Four Shock Isolators

(3) Six electrical connectors

3.5.14.5 Controls and Indicators - The Base shall not contain any controls or indicators.

3.5.14.6 Electrical Confections - The electrical connections shall be in accordance with Figure 19.

3.5.15 Base, Shockmount, Electrical Equipment MT-4617/ARN-84(V) - The Base shall meet the applicable requirements of MIL-C-172.

3.5.15.1 Function - The Base shall provide suitable mounting provisions for the Receiver-Trasmitter RT-1127/ARN-84(V) and Signal Data Converter CV-3136/ARN-84(V), and shall provide for electrical connections.

3.5.15.2 Form Factor - The form factor of the Base shall be in accordance with Figure 20.

3.5.15.3 Weight - The weight of the Base shall not exceed 8.0 pounds.

3.5.15.4 Contents - The Base shall contain the following assemblies:

(1) Mounting Base

(2) Eight Shock Isolators

3.5.15.5 Controls and Indicators - The Base shall not contain any controls or indicators.

3.5.15.6 Electrical Connections - The Base shall not have any electrical connections.

3.6 Design Data - No data is required by this specification (other than reports accompanying samples submitted for testing) or by applicable documents, unless specified in the contract or order.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplier and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction (First Article) Tests

(2) Initial Production Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction (First Article) Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure. (See 4.6) The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422.

4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the

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approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

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4.3 Initial Production Tests - One of the first 10 production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment

4.3.3 Initial Production Sample Approval - Approval of the Initial Production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractor not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or

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approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjetted to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run-In Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run-In Test - Each equipment shall be operated under the conditions specified herein for a period of 10 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment. This test shall be deleted if the reliability test includes a test on each equipment which consumes at least 10 hours of operation.

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Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 2 g's.

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 10-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment is 2 directions the vertical direction shall be used. During the 10-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 10-hour period specified above may be composed of two 5-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Assurance Tests are deleted by contractual action.)

Quantity of Equipment Offered for Acceptance	Quantity to be Selected for Sampling
First 10	0*
Next 10	1
Next 50	1
Next 50	1
1 for each additional 200 or fraction thereof	

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*NOTE: If by contract action the Initial Production Test is deleted, then a Sample Test shall be conducted on the equipment from the first 10 produced.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be sub jetted to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational tests at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, Initial production, individual and special tests.

(3) Manufacturing run-In test, Specified in 4.4.1.3, except that the test duration shall be 50 hours- with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in fixture equipment.

4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as requlred by both the Qualification Phase and the Production Acceptance (Sampling) Phase shall be conducted. Classification of failure shall be in accordance with MIL-STD-781 and AR-34.

4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall. be those listed In Table 5 of MIL-STD-78l. For the Qualification Phase, Test Level F shall be used. The Accept-Reject Criteria for Test Plan II shall be used. External forced air cooling air at 71°C and 0.95 pounds/min. shall be used during the heating portion of the temperature cycle. During the cooling portion of the temperature cycle, forced air cooling shall not be used.

4.4.3.2 Reliability Production Acceptance (Sampling) Phase Tests - The equipment, throughout production, shall be tested as outlined in MIL-STD-78l (as modified herein) under the section entitled "Production Acceptance (Sampling) Phase of Production Reliability Tests." Test level F shall be used. External forced air cooling at 71°C and 0.95 lbs/min. shall be used during the heating portion of the temperature cycle. During the cooling port ion of the temperature cycle, forced air cooling shall not be used.

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4.4.3.2.1 All Equipment Test - Each Equipment produced except those submitted for the Reliability Qualification Test, shall be tested for 50 hours l Prior to the 50 hour test on each equipment, a burn-in period may be used at the option of the contractor. If the burn-in period is to be used the details thereof must be included In the approved test procedures. To determine whether the MTBF is being met at any time during the contract the operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan III of MIL-STD-781. (Extend the line as necessary to accomdate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attention: Avionics Division. At any time that the current totals of test hours and test failures slotted on Test Plan III curves show a reject situation, the procuring activity shall be notified. Notwithstanding the fact that as data are accumulated the cumulative plot based upon such actual data would pass into the "accept area" of the graph Illustrated in Test Plan III of MIL-STD-781, the cumulative plot of failures versus test time shall not be plotted below or to the right of the "accept line" of that graph and the lower or "accept line" shall for this purpose act as a limiting boundary of the plot. "Accept" criteria for initial and subsequent production deliveries is considered having been met based on successful completion of the Reliablity Qualification Test required prior to Initial delivery of the equipment and the absence of a "reject" decision. At any time that the current totals of test hours and test failures plotted on Test Plan III curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision.

4.4.3.3 Test Details - The test details such as the length of the test cycle, the performancee characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the tests, etc., shall be part of the test procedures to be subitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests. The heat portion of the test cycle shall not exceed four (4) hours.

4.4.4 Special Tests- Special tests shall be concluded on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

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4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. This will be determined by procuring activity).

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contract shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, prposed corrective action intended to reduce the possibility of same failure(s) occuring in future tests.

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(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the Government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) when reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Periods - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal Room
Altitude	Normal Ground (0-5000 ft)
Humidity	Room Ambient
A.C. Voltage	115 ± 5 volts (at lowest applicable frequency)
DC Voltage	27.5 ± 2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment-temperature to stabilize.

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Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hours intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the Government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning Of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all “wear” items. The “wear” items shall be determined by

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agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government inspector.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specifications MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in Specification MIL-E-17555, the method Of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

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6. NOTES

6.1 Intended Use - The equipment provides distance and bearing information to an aircraft pilot. It is used in fixed and rotarv-wins aircraft, and the information is with reference to a land-, or carrier-based, beacon, or another similarly equipped aircraft.

6.2 Ordering Data - Purchasers should exercise any desire options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (see 5.1).

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. all referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this

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equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 Non-Repairable Subassemblies - As a general rule non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.6 Type Designations - The parentheses (*), when used in the type de be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with Specification MIL-N-18307. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.7 Associated Equipment - The equipment shall operate with the following associated equipment, or their equivalents, which shall not be supplied as part of the unit.

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Indicator, Bearing - Distance Heading ID-663/U	(MIL-I-22075)
Indicator, Course ID-387/ARN	(MIL-I-9229)
Indicator, Course ID-250A/ARN	(MIL-C-5824)
Indicator, Range ID-388/APN	(MIL-I-18955)
Indicator, Horizontal Situation ID-1013/A	(MIL-I-23366)
Indicator, Horizontal Situation AQU-4A	(MIL-I-27848)
Switch, Radio Frequency Transmission Line SA-521/A	(MIL-S-25879)
Antenna AT-741/A (or electrical equivalent)	MIL-A-25708)
Data Link Set AN/ASW-27(*)	(MIL-D-81170)

6.8 Configurations - The various configurations for the AN/ARN-84 TACAN Navigational Set are listed in Table IV.

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TABLE I. Operation Vs. Function

Function Selector Position	OFF	REC	T/R	A/A
	No Operation	Receive ID Tone and Bearing from a beacon with modulating antenna	Same as REC function plus Range	Receive/Transmit slant range to/from a complementary aircraft. Receive and display Bearing to an aircraft transmitting modulated signals.

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TABLE III. Operation Vs. Function

Function Selector Position	OFF	REC	T/R	A/A	BCN
Normal	No Operation	Same As Table I	Same As Table I	Same As Table I	Receive/Transmit slant range to/from a complementary aircraft. Receive Bearing from a complementary aircraft transmitting modulated signals. Transmit squitter pulses.
Inverse	No Operation	Receive ID Tone and Bearing from a DME beacon with an omni antenna	Same as Inverse REC function plus Range	Receive/Transmit slant range to/from a complementary aircraft. Receive Bearing from a complementary aircraft transmitting squitter pulses.	Receive/Transmit slant range to/from a complementary aircraft. Receive Bearing from a complementary aircraft transmitting squitter pulses. Transmit squitter pulses and reference bursts.

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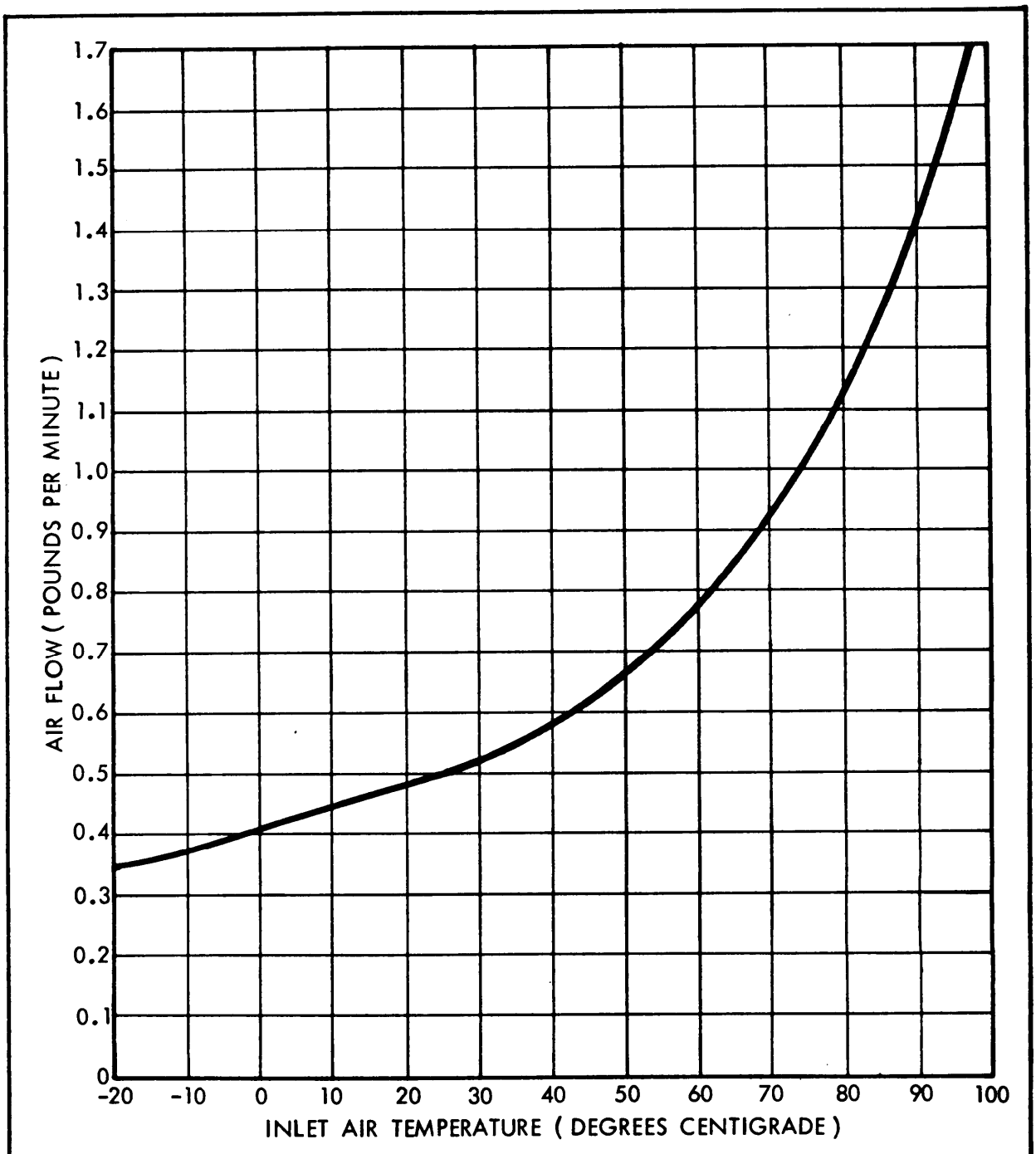
TABLE IV. CONFIGURATION MATRIX

(V)1	28V, Red Light Control With Soft Mount	(V)2	28V, Red Light Control With Hard Mount	(V)3		(V)4	5V, Red Light Control With Soft Mount
RT-1022/ARN-84(V)		RT-1022/ARN-84(V)		RT-1022/ARN084(V)		RT-1022/ARN-84(V)	
CV-2837/ARN-84(V)		CV-2837/ARN-84(V)		CV-2837/ARN-84(V)		CV-2837/ARN-84(V)	
C-8734/ARN-84(V)		C-8734/ARN-84(V)		- - -		C-9054/ARN-84(V)	
MT-4354/ARN-84(V)		MT-4443/ARN-84(V)		- - -		MT-4354/ARN-84(V)	

(V)5	5V, Red Light Control With Hard Mount	(V)7	AN/ARN-21 Retrofit With 5V, White Light Control	(V)8	AN/ARN-52 Retrofit With 5V, White Light Control	(V)9	AN/ARN-65 Retrofit With 5V, White Light Control
RT-1022/ARN-84(V)		RT-1127/ARN-84(V)		RT-1127/ARN-84(V)		RT-1127/ARN-84(V)	
CV-2837/ARN-84(V)		CV-3134/ARN-84(V)		CV-3135/ARN-84(V)		CV-3136/ARN-84(V)	
C-9054/ARN-84(V)		C-9475/ARN-84(V)		C-9475/ARN-84(V)		C-9475/ARN-84(V)	
MT-4443/ARN-84(V)		MT-4615/ARN-84(V)		MT-4616/ARN-84(V)		MT-4617/ARN-84(V)	

(V)16	AN/ARN-21 Retrofit With 5V, Red Light Control	(V)17	AN/ARN-52 Retrofit With 5V, Red Light Control	(V)18	AN/ARN-65 Retrofit With 5V, Red Light Control		
RT-1127/ARN-84(V)		RT-1127/ARN-84(V)		RT-1127/ARN-84(V)			
CV-3134/ARN-84(V)		CV-3135/ARN-84(V)		CV-3136/ARN-84(V)			
C-9483/ARN-84(V)		C-9483/ARN-84(V)		C-9483/ARN-84(V)			
MT-4615/ARN-84(V)		MT-4616/ARN-84(V)		MT-4617/ARN-84(V)			

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NAVIGATIONAL SET ,TACAN AN/ARN-84 COOLING AIR REQUIREMENTS	DRAWING NO. Figure 1	REV.

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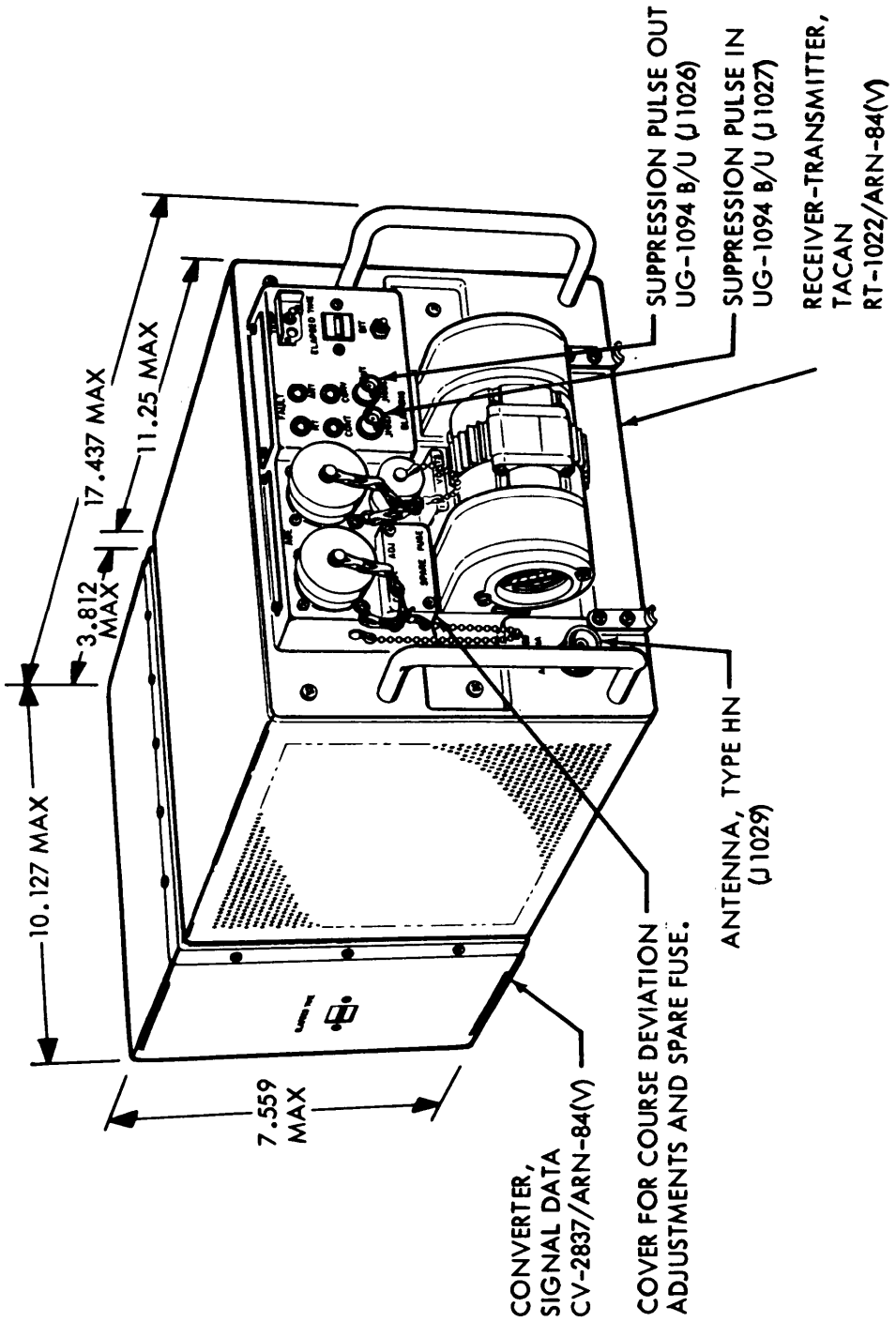
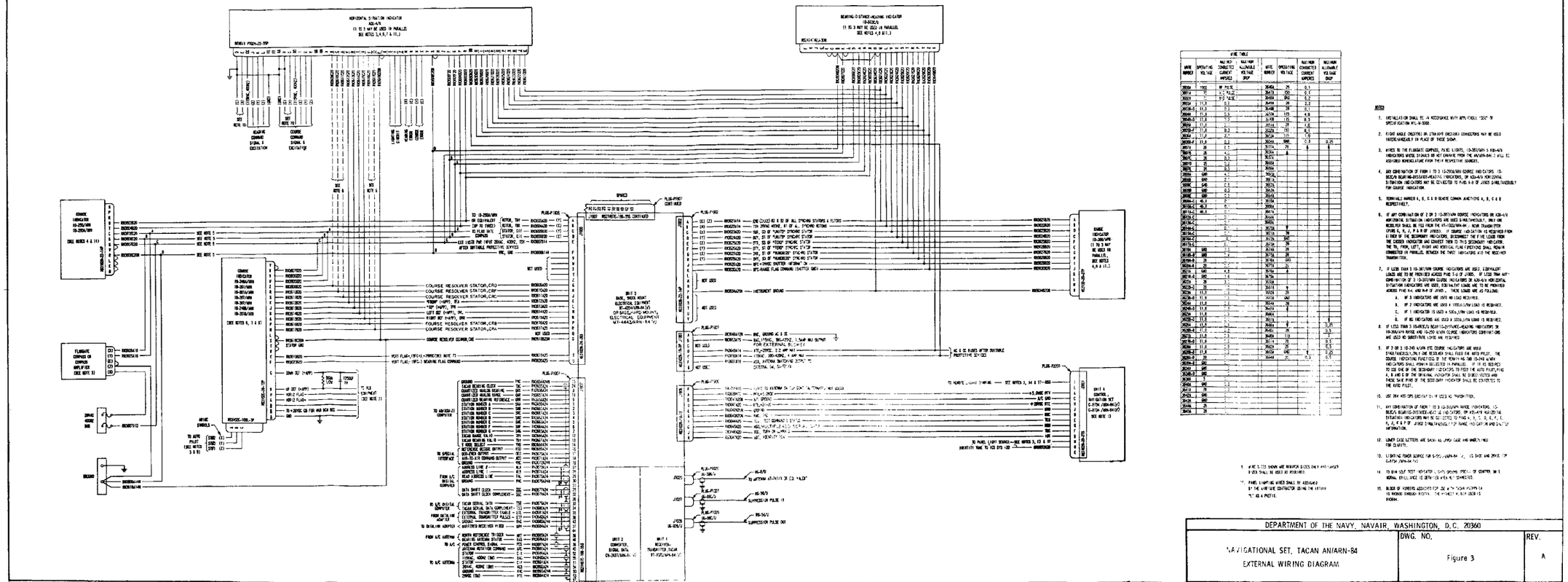


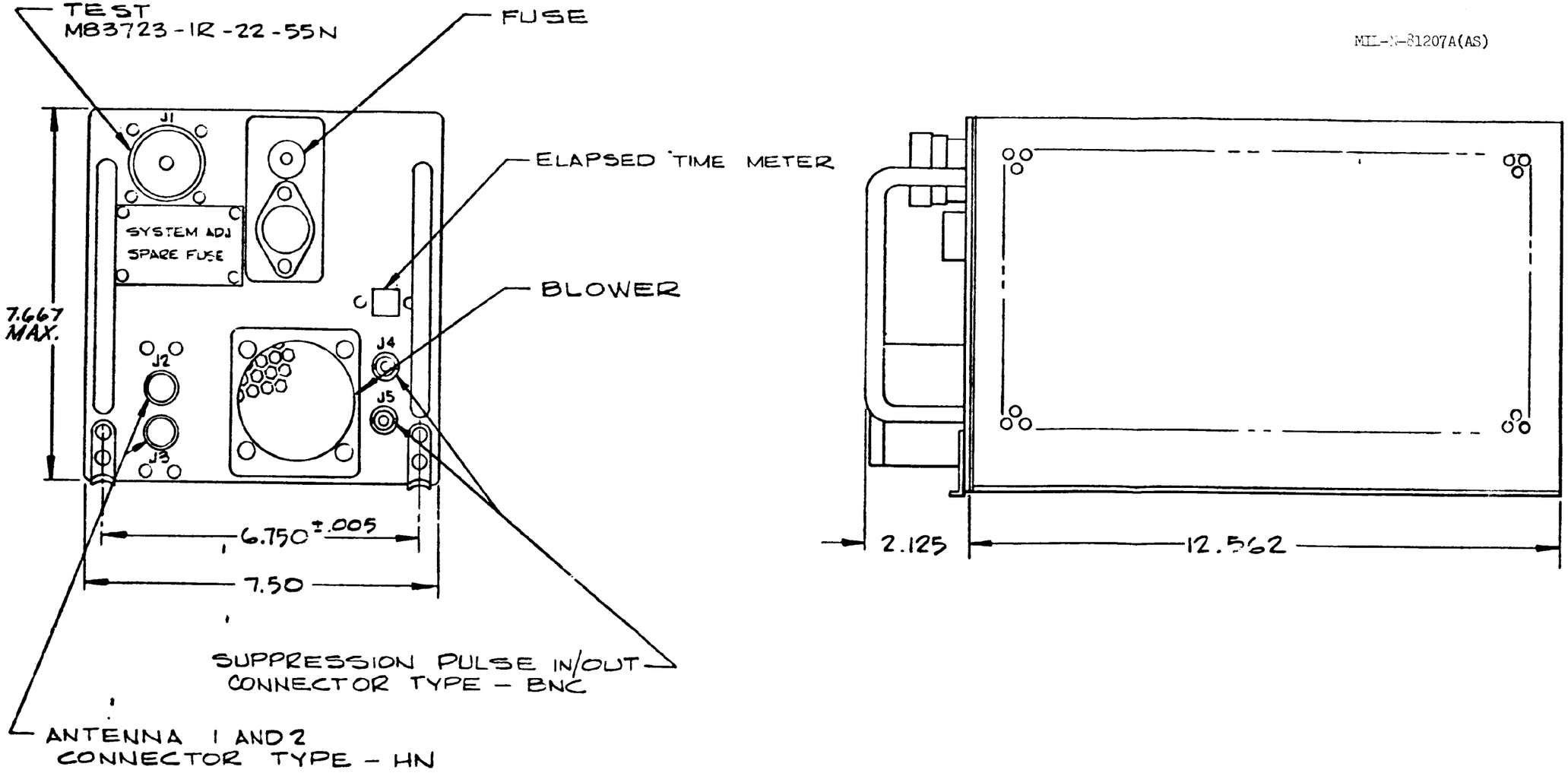
Figure 2. FORM FACTOR RT-1022/ARN-84(V) AND CV-2837/ARN-84(V)

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NAVIGATIONAL SET, TACAN AN/ARN-84 EXTERNAL WIRING DIAGRAM	DWG. NO. Figure 3	REV. A

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Figure 4 Form Factor RT-1127/ARN-84(V)

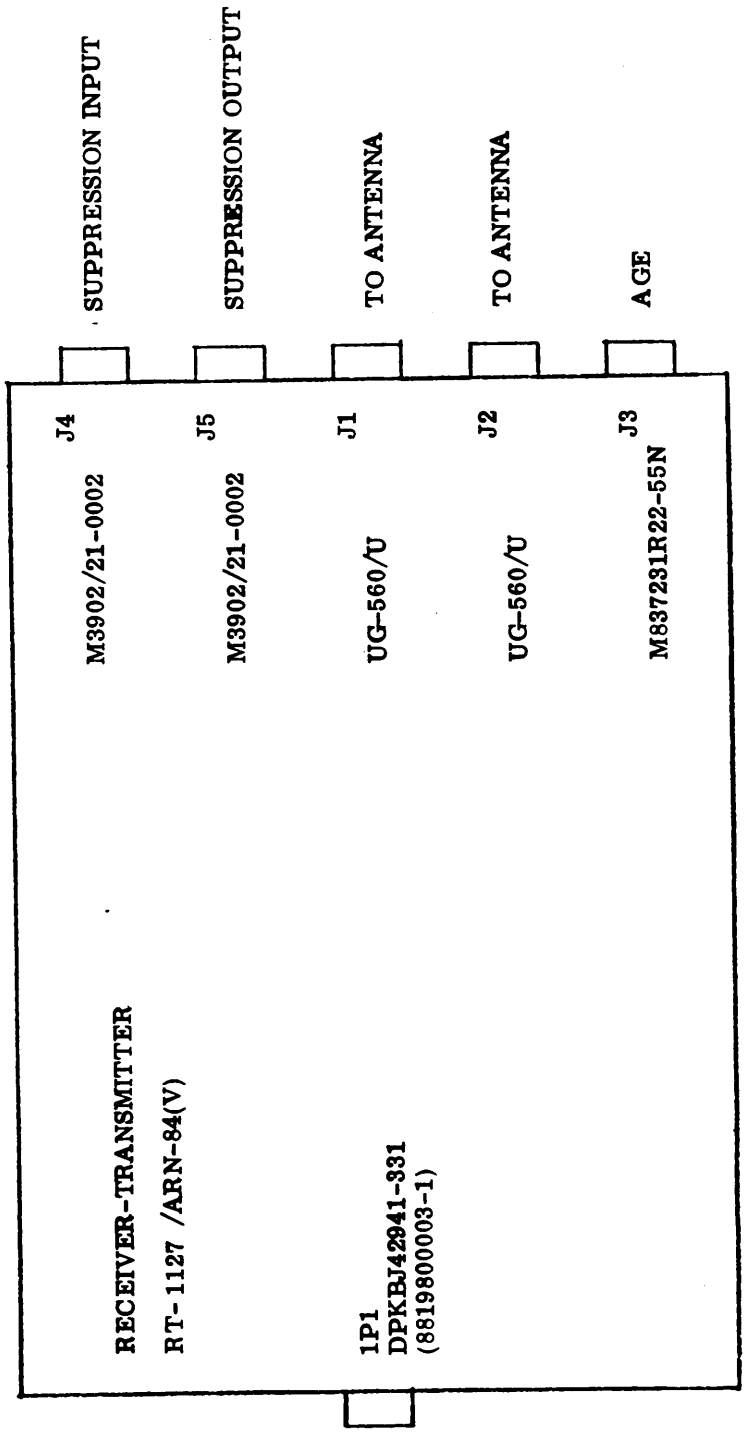


FIGURE 5. ELECTRICAL CONNECTIONS RT-1127/ARN-84(V)

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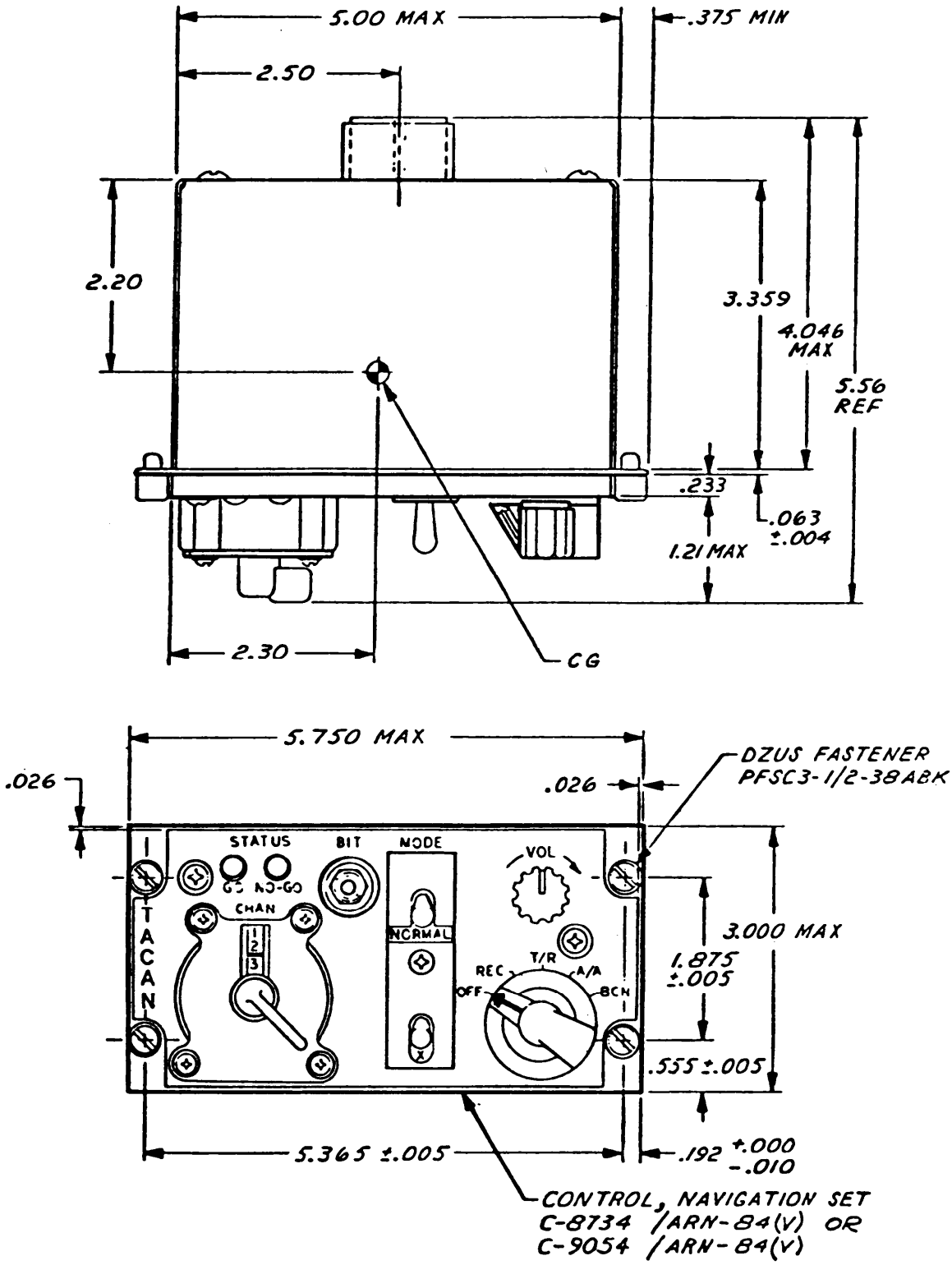


Figure 6. Form Factor Control Unit

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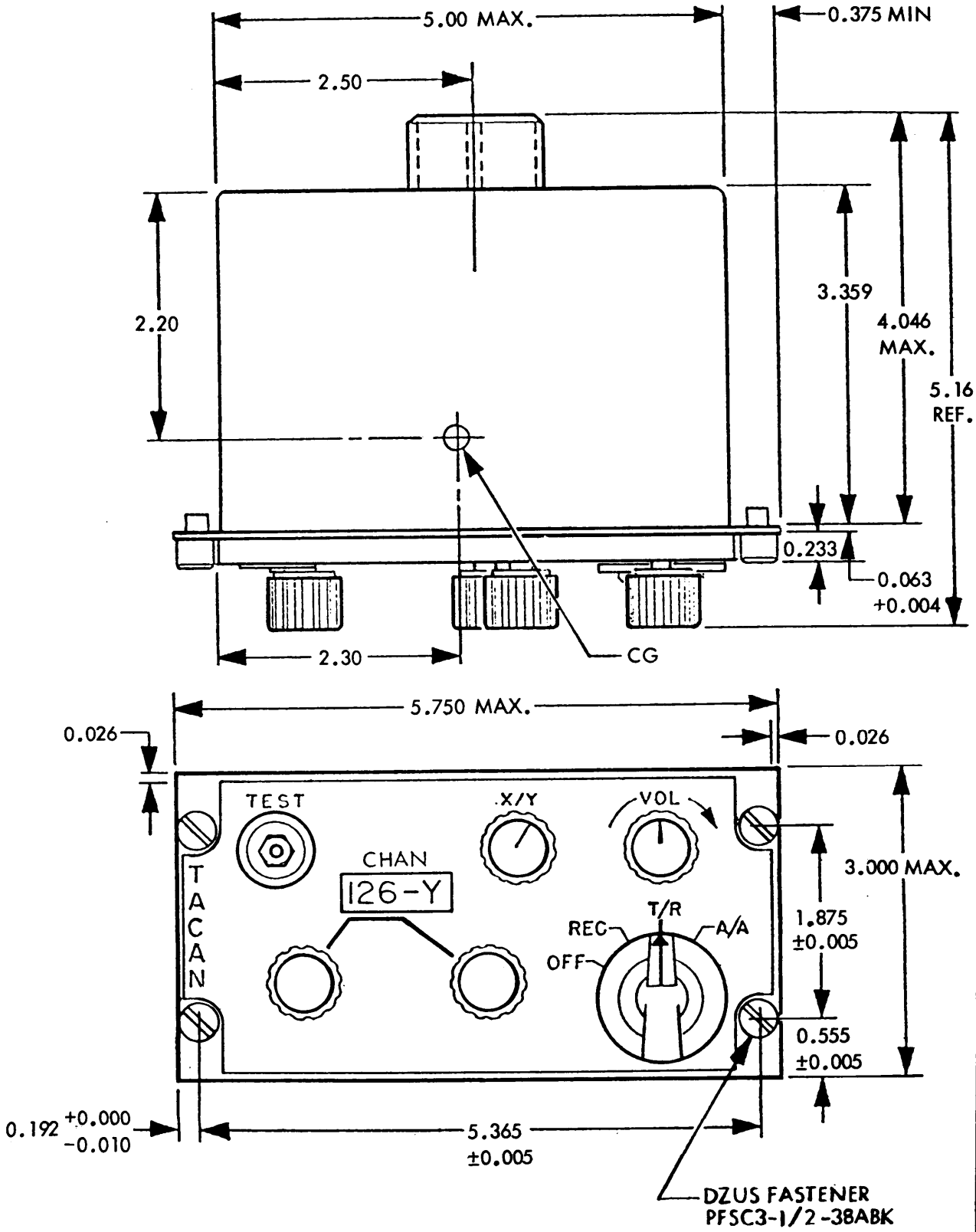


Figure 7. Form Factor, Navigation Set Control C-9475/ARN-84(V) or C-9483/ARN-84(V)

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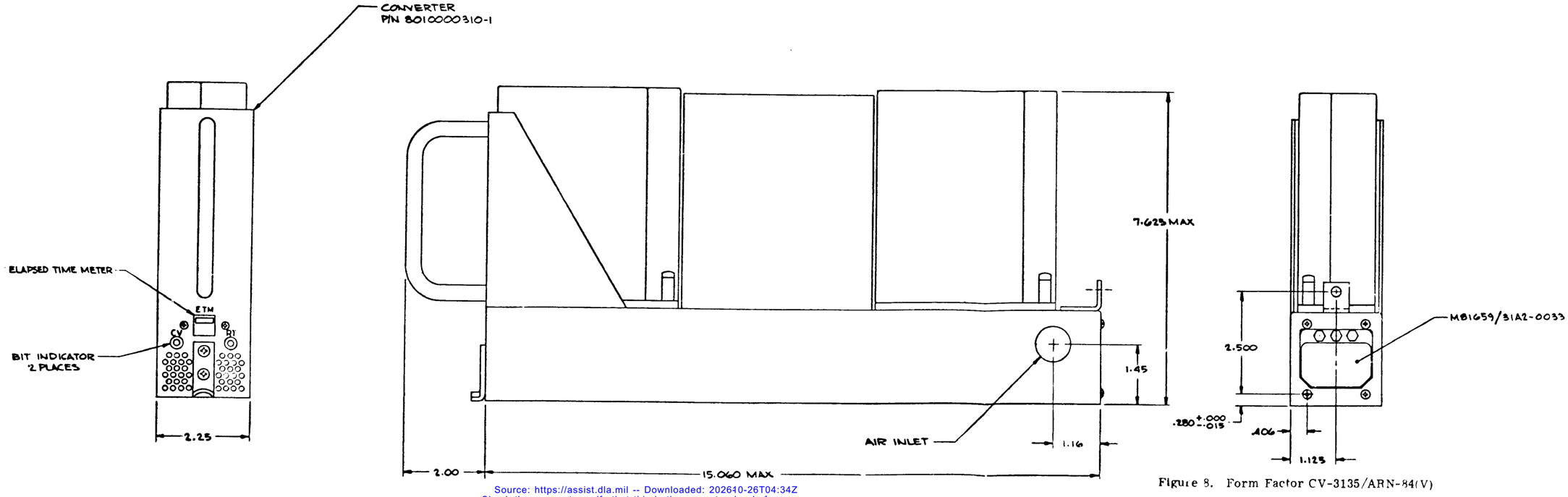


Figure 8. Form Factor CV-3135/ARN-84(V)

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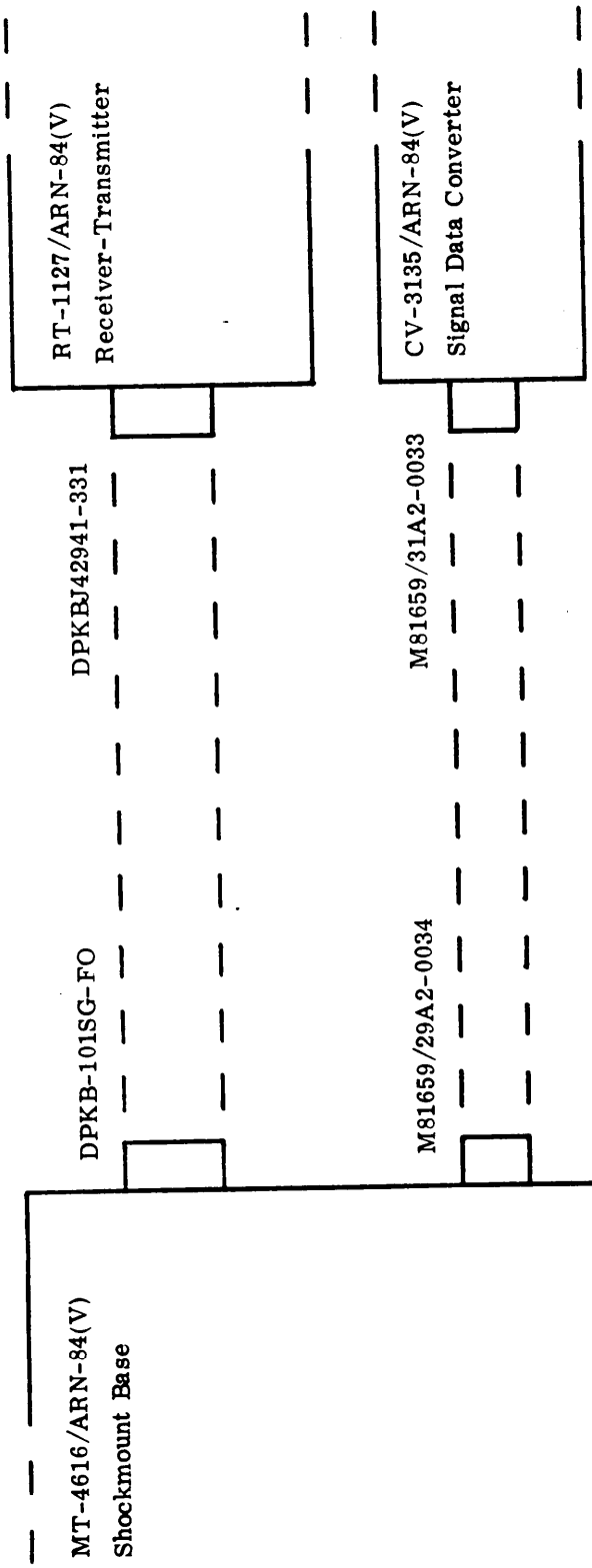


Figure 9. Signal Data Converter CV-3135/ARN-84(V) Electrical Connections

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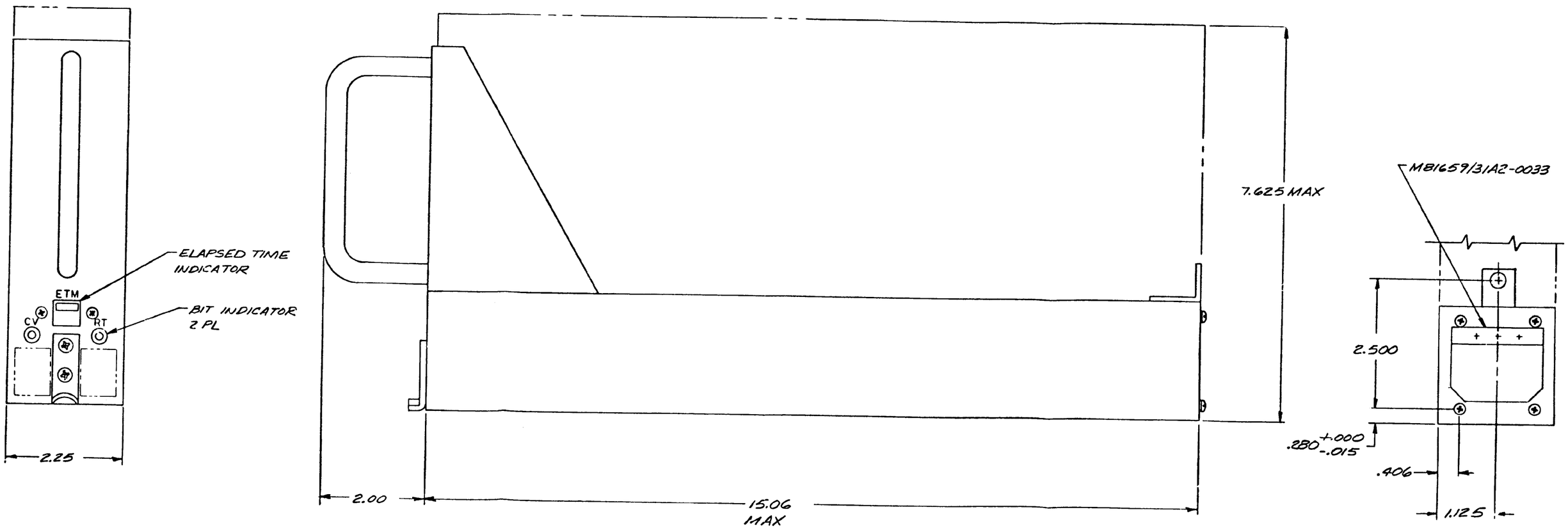


Figure 10. Form Factor CV-3134/ARN-84(V)

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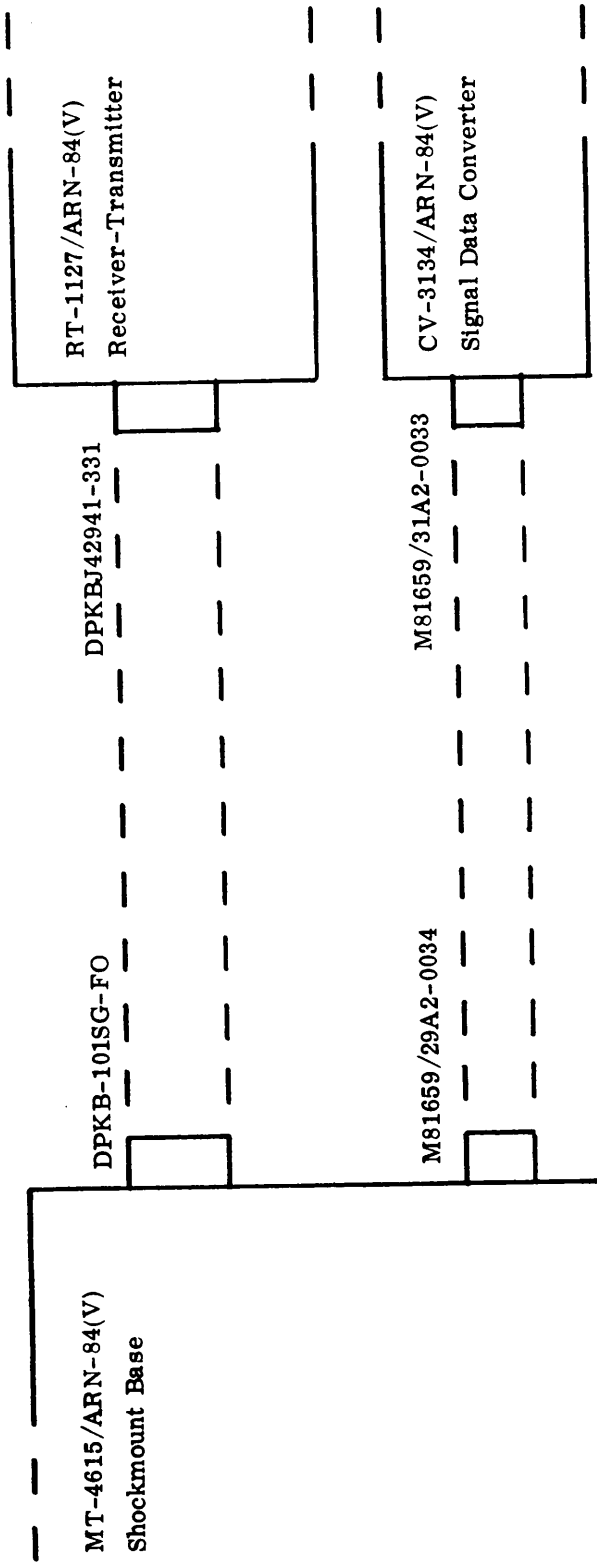
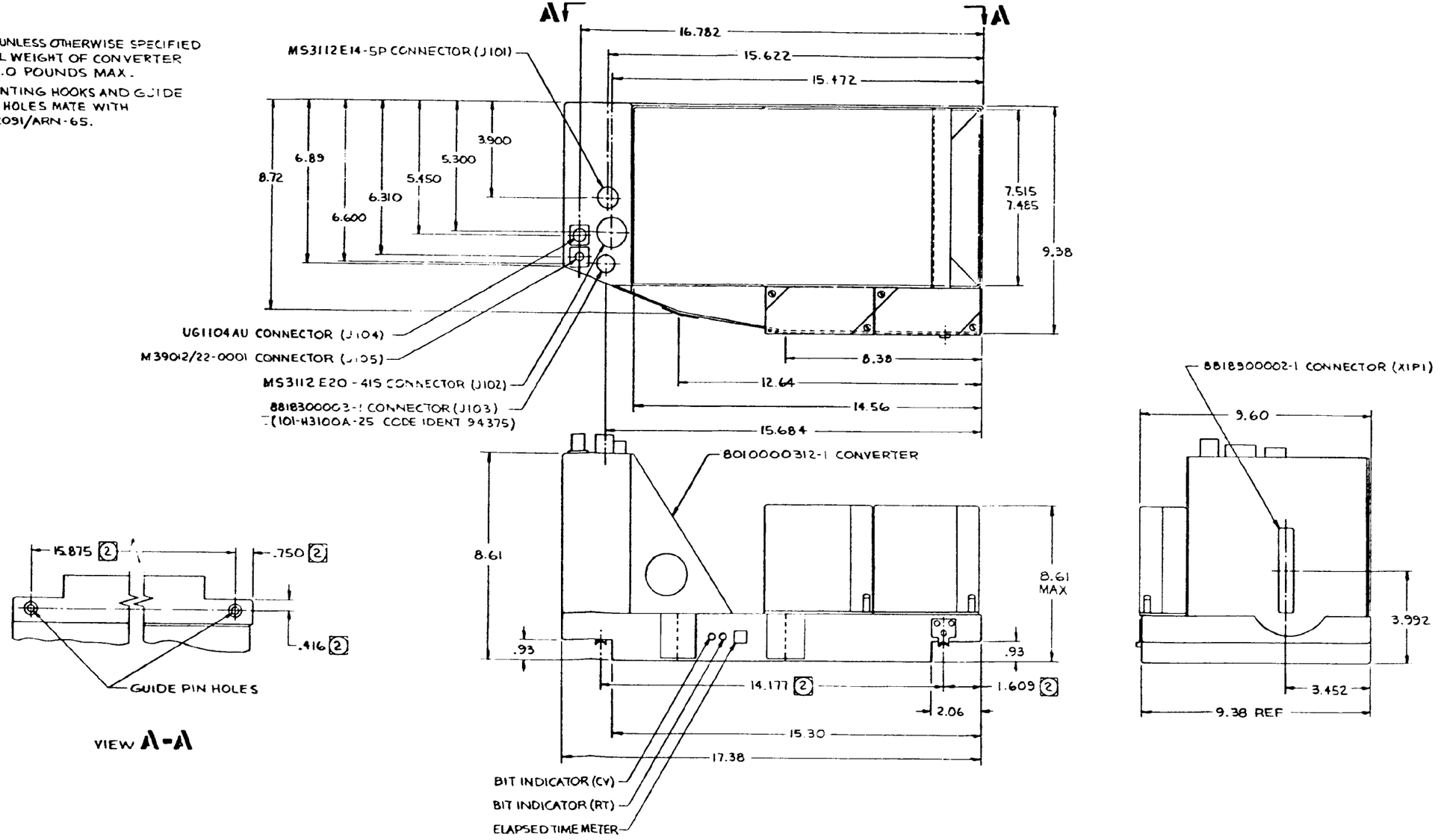


Figure 11. Signal Data Converter CV-3134/ARN-84(V) Electrical Connections

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NOTES: UNLESS OTHERWISE SPECIFIED
1. TOTAL WEIGHT OF CONVERTER IS 13.0 POUNDS MAX.
2. MOUNTING HOOKS AND GUIDE PIN HOLES MATE WITH MT 2091/ARN-65.



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Figure 12. Form Factor CV-3136/ARN-84(V)

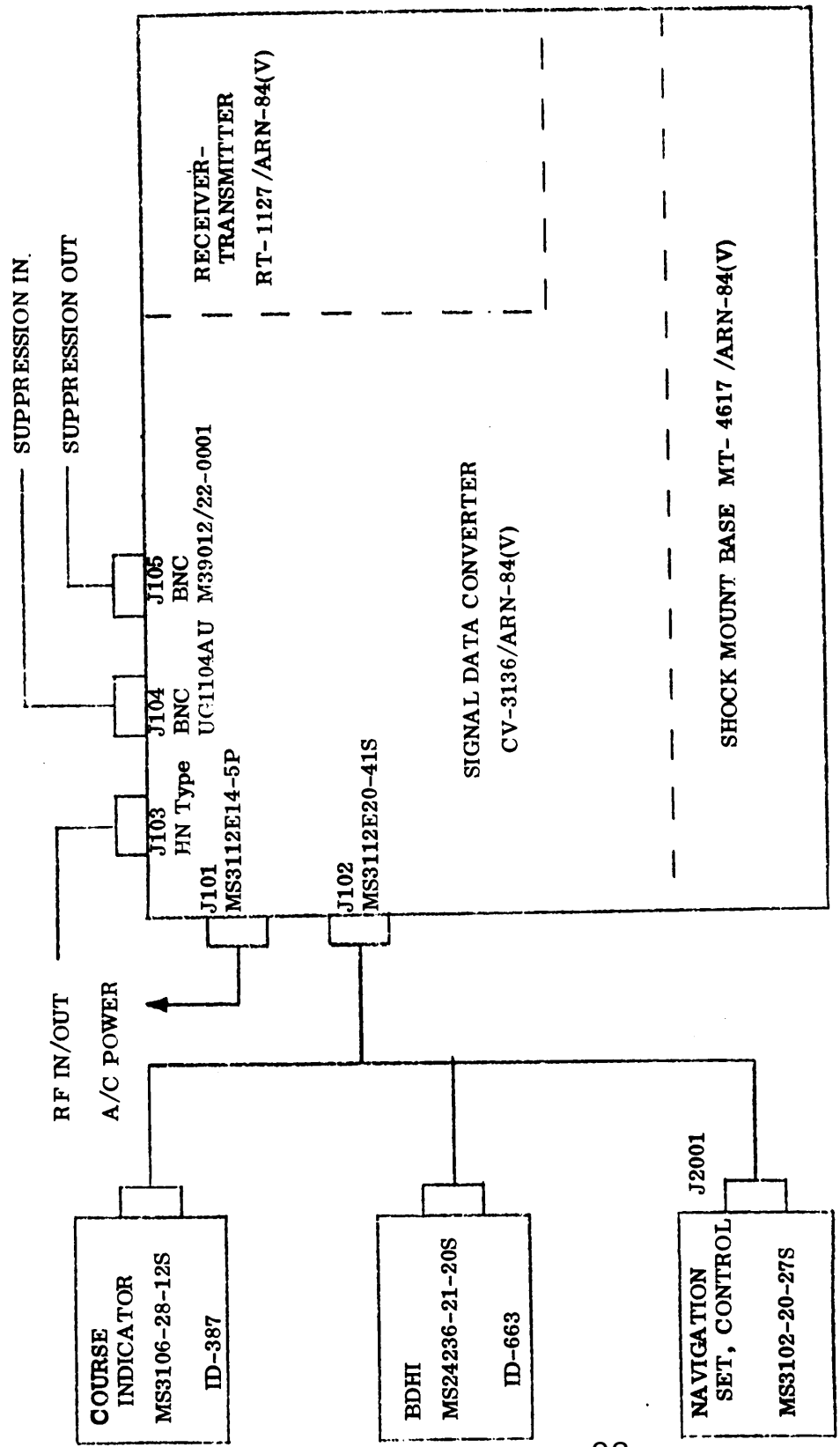


FIGURE 13. WIRING DIAGRAM CV-3136/ARN-84(V)

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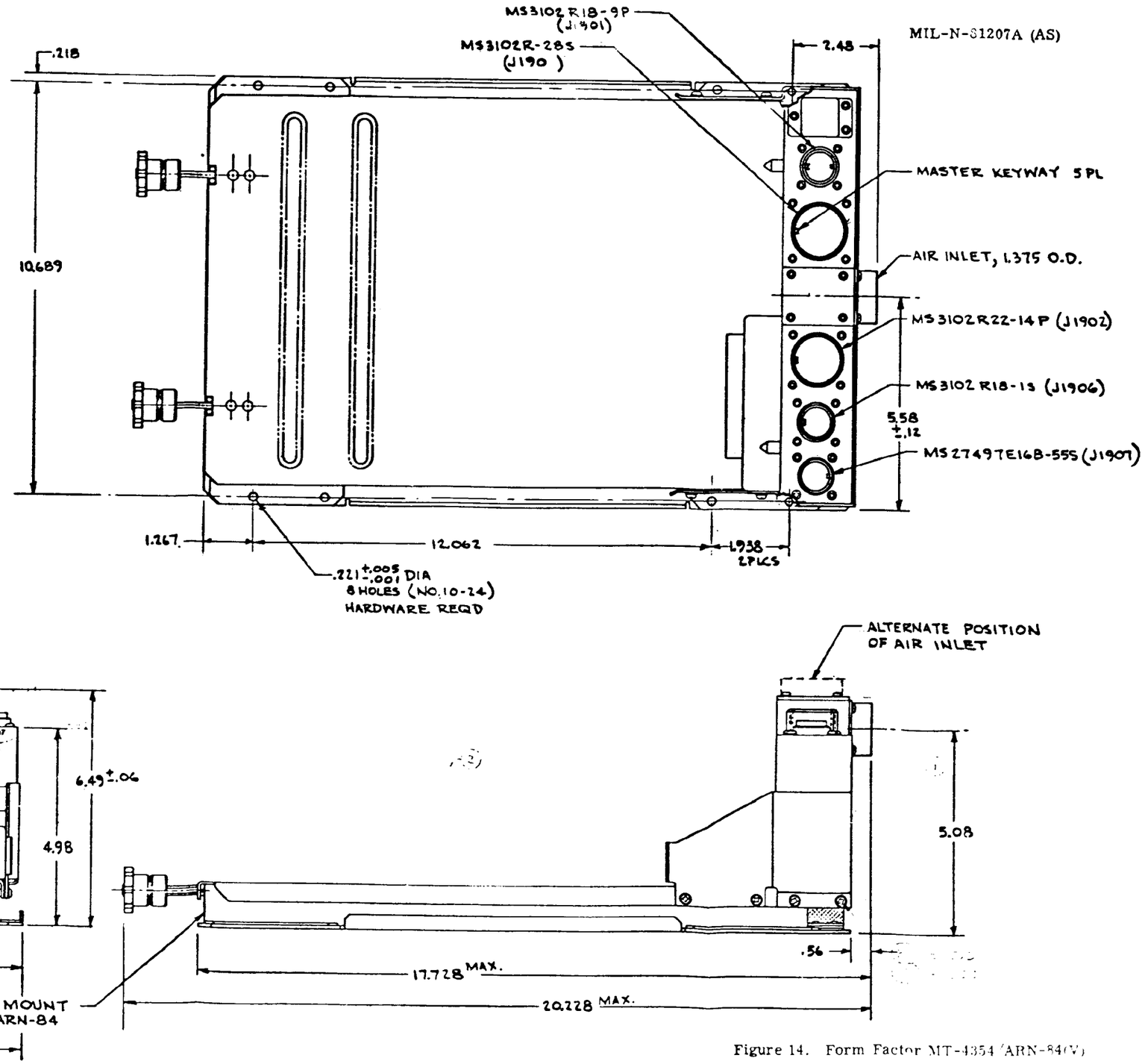


Figure 14. Form Factor MT-4354 (ARN-84(V))

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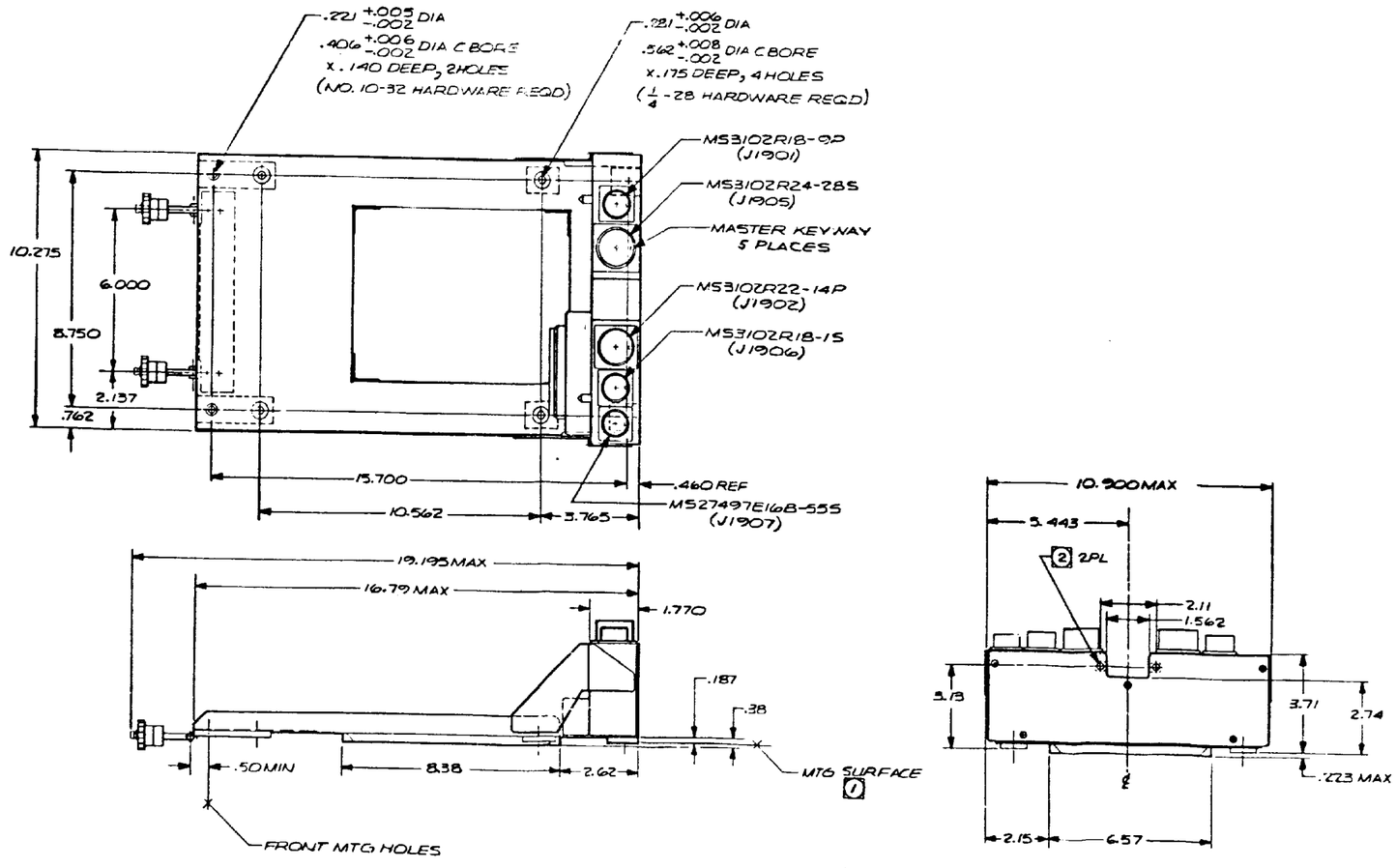
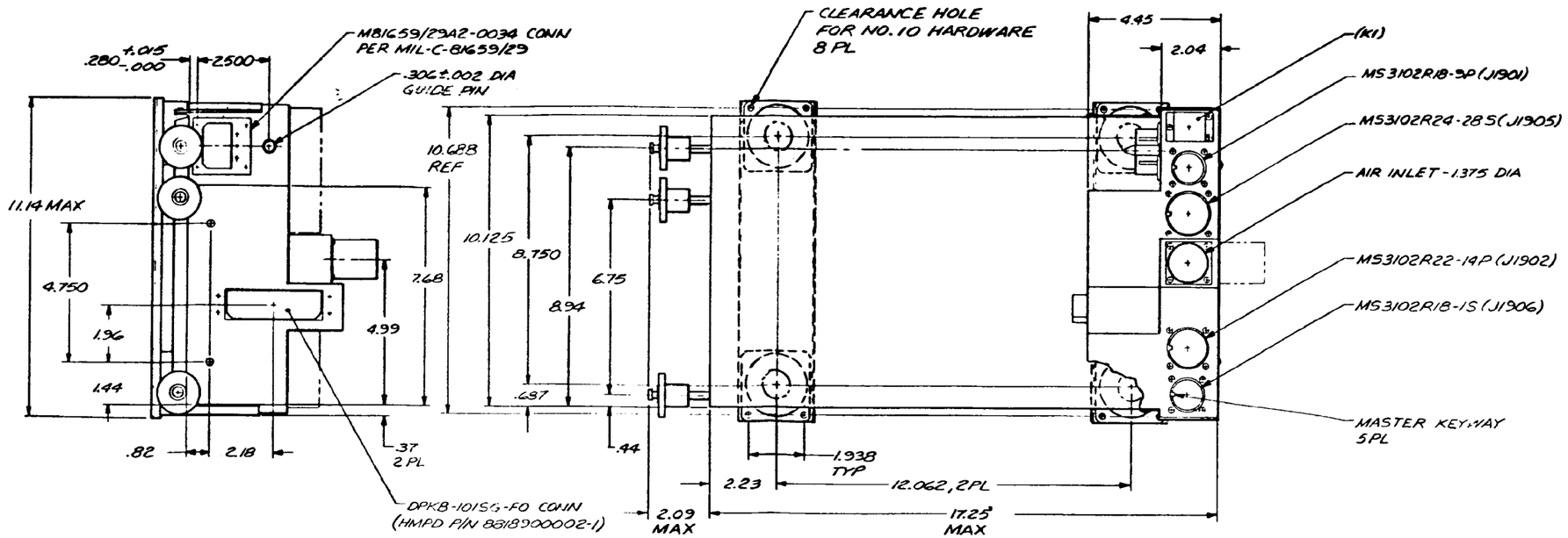


Figure 15. Form Factor MT-4443/ARN-84(V)

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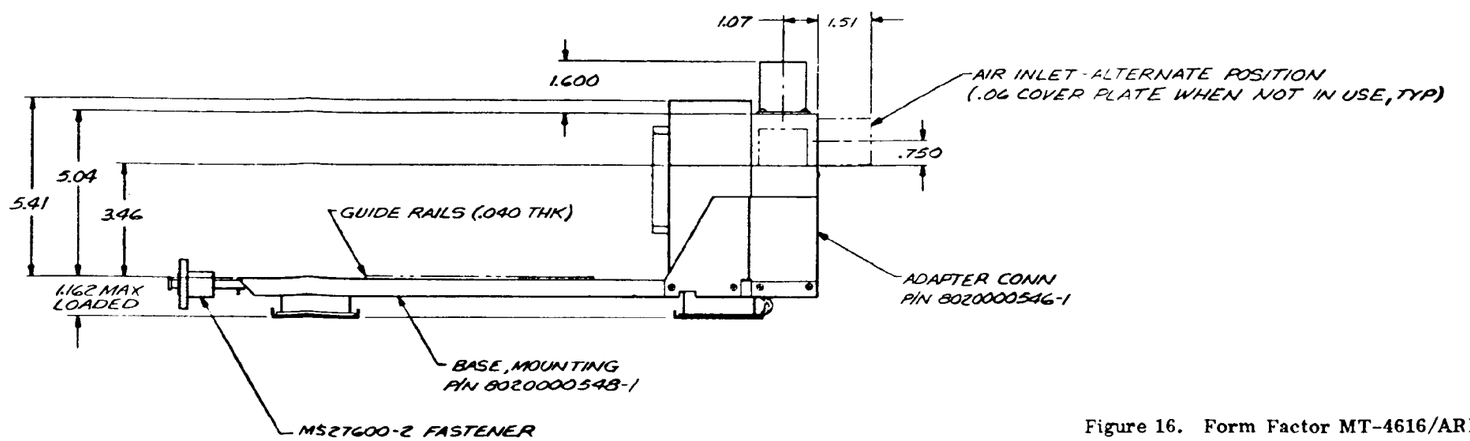


Figure 16. Form Factor MT-4616/ARN-84(V)

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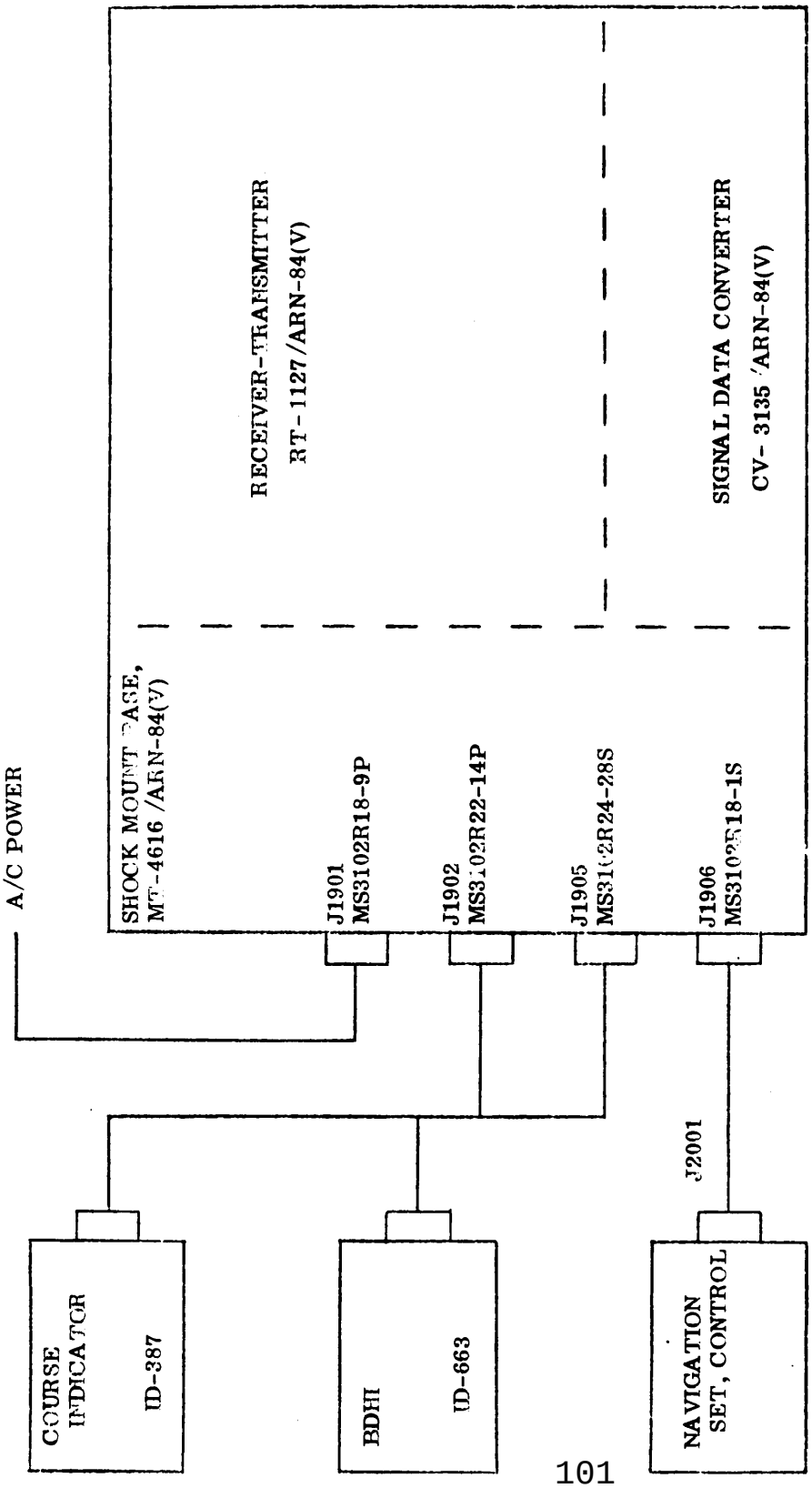


FIGURE 17. AIRCRAFT WIRING DIAGRAM MT-4616/ARN-84(V)

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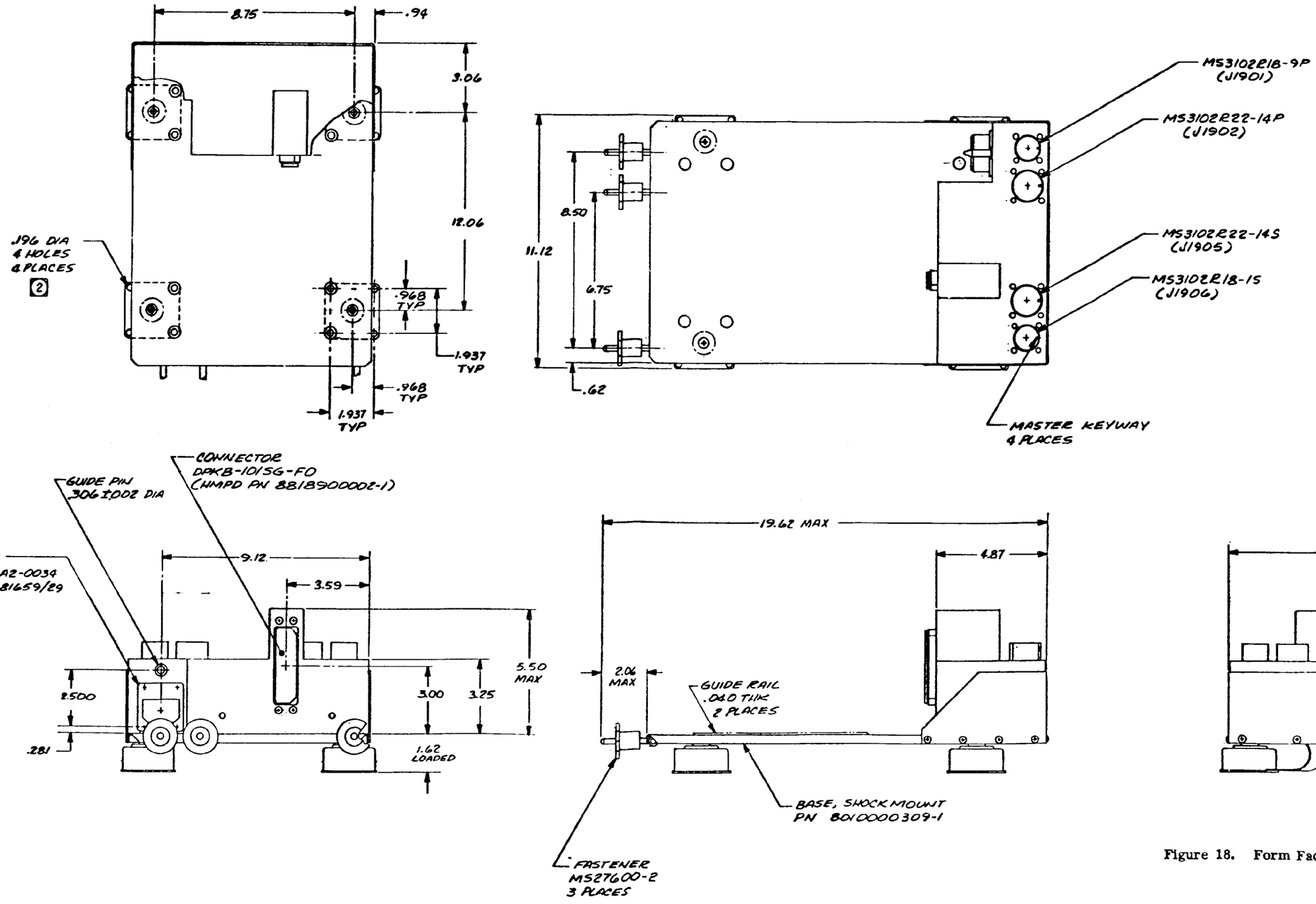


Figure 18. Form Factor MT-4615/ARN-84(V)

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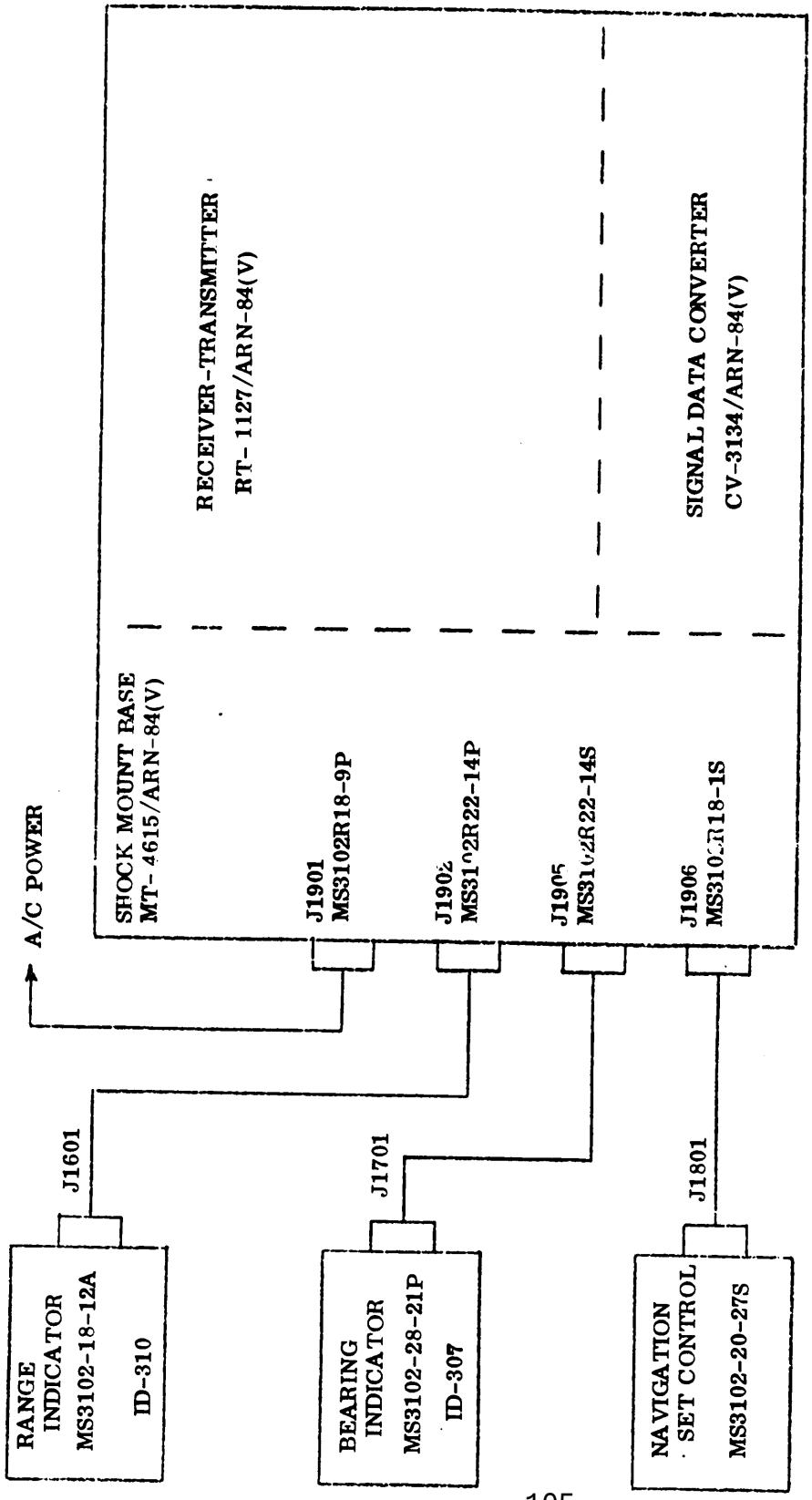
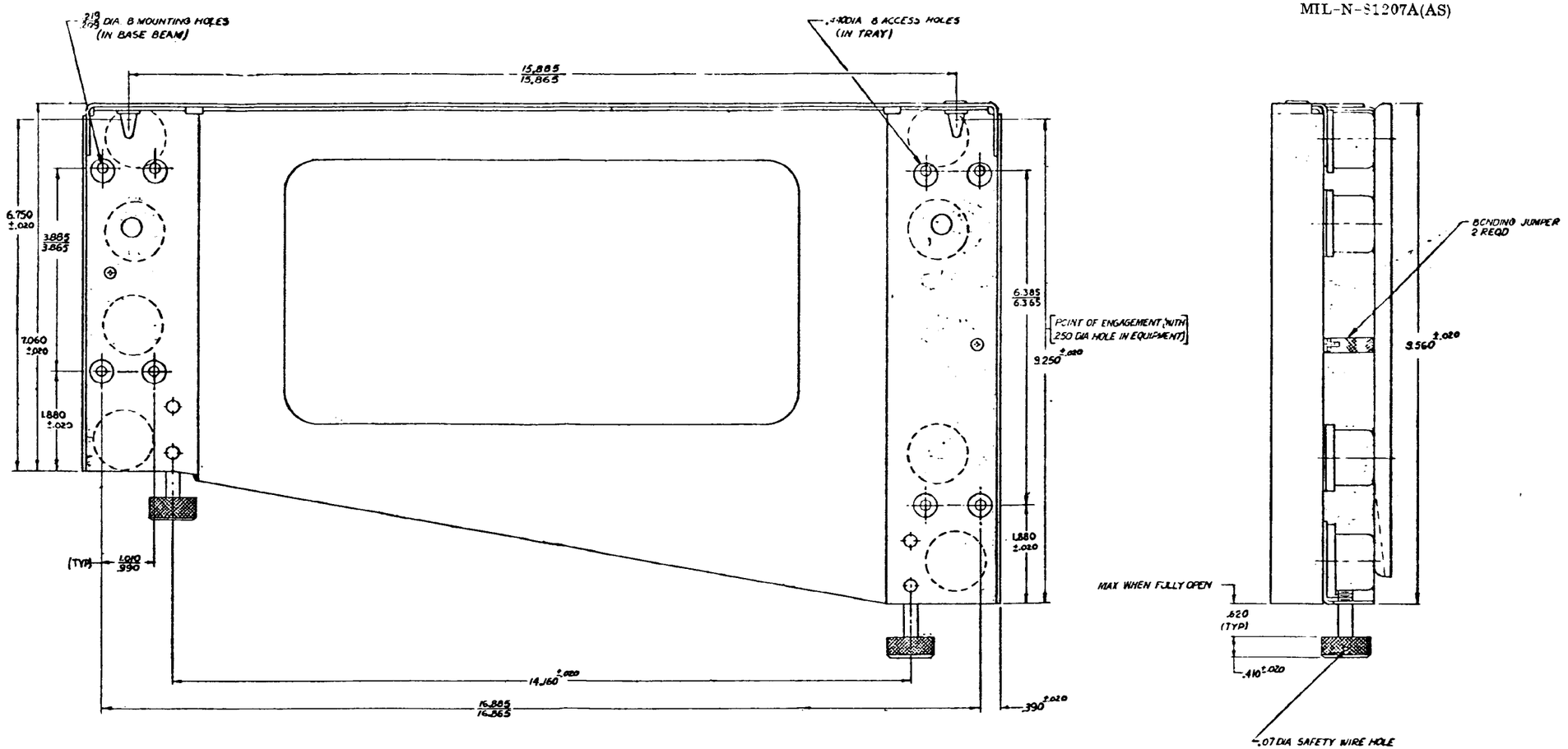


FIGURE 19. AIRCRAFT WIRING DIAGRAM MT-4615/ARN-84(V)

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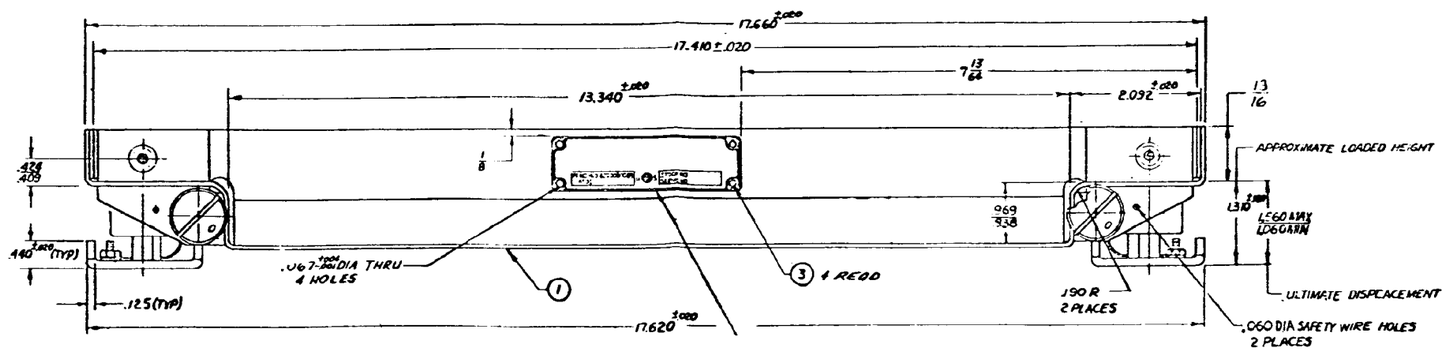


Figure 20. Form Factor MT-4617/ARN-84(V)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-N-81207A NAVIGATIONAL SET, TACAN AN/ARN-84(V)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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Naval Air Systems Command
Washington, D.C. 20361

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